AN INTELLIGENT AI-DRIVEN DECISION SUPPORT SYSTEM FOR PERSONALIZED INCOME TAX COMPLIANCE AND ADVISORY IN SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Understanding and complying with personal income tax requirements in Sri Lanka can be challenging because taxpayers must interpret financial transactions, apply the correct tax legislation, understand relevant legal provisions, and identify strategies suited to their financial circumstances. To address these challenges, this research project presents an intelligent AI-driven decision support system for personalized income tax compliance and advisory in Sri Lanka. The proposed integrated solution consists of four components: Transaction Semantic Reasoning, which interprets and categorises financial transactions; Optimization and Explainable Engine, which applies year-specific tax rules and performs deterministic tax calculations with supporting explanations; Intelligent Tax Advisory Language Model, which provides natural-language responses grounded in relevant tax information; and Personalized Recommendation and Predictive Impact Modeling Engine, which ranks potentially suitable tax strategies and estimates their possible financial impacts. The overall methodology combines tax legislation analysis, rule-based processing, artificial intelligence, information retrieval, machine-learning-based prediction, and explainability techniques. The components are designed to operate within a shared application while maintaining clear distinctions between legally defined tax calculations and model-generated predictions. The integrated approach aims to connect financial information processing, tax assessment, advisory support, and personalised recommendations within a consistent workflow. The system provides a research-based approach to making tax-related information more understandable and supporting users in reviewing possible financial decisions. However, the reliability and practical effectiveness of the complete solution depend on the accuracy of the underlying tax rules, the quality of the information and models, and the results of systematic testing and validation. Overall, the project demonstrates an integrated approach to AI-assisted personal income tax decision support in Sri Lanka, with potential value for improving the accessibility, traceability, and personalisation of tax-related information.

Keywords: Personal Income Tax, Artificial Intelligence, Explainable AI, Tax Advisory, Decision Support System

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
GDPR	General Data Protection Regulation
IRD	Inland Revenue Department
ML	Machine Learning
NLP	Natural Language Processing
OECD	Organisation for Economic Co-operation and Development
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
SDGs	Sustainable Development Goals
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
UX	User Experience

Table 1: List of Abbreviations Used in the Report

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

1.1.1 Personal income tax in Sri Lanka

Personal income tax is an important part of Sri Lanka's financial and regulatory system. Individuals need to identify their taxable income, understand which tax reliefs may apply to them, and calculate their tax liabilities according to the legislation relevant to the correct year of assessment. The Inland Revenue Act, No. 24 of 2017, together with its subsequent amendments, provides the main legal framework for income tax in Sri Lanka [1].

Understanding these requirements can be challenging because tax rules may change over time. Legislative amendments can introduce new provisions or revise tax rates, relief limits, and eligibility conditions. Therefore, taxpayers must identify the rules that apply to their individual circumstances and the relevant year of assessment. Accurate tax assessment requires both reliable financial information and the correct interpretation of the applicable legal provisions [1].

These requirements can be particularly difficult for individuals who have limited knowledge of taxation or cannot regularly access professional tax advice. Completing a tax assessment involves more than entering income into a calculator. Individuals may also need to interpret financial records, understand legal terminology, identify eligible reliefs, and maintain supporting information for their declarations. Digital tax administration initiatives demonstrate the growing importance of technology in making tax-related processes more accessible and efficient [2], [3].

1.1.2 Challenges in existing tax assessment and advisory practices

Although online tax calculators and information sources can help taxpayers understand their basic obligations, calculating a tax amount alone does not necessarily explain how that amount was obtained. Taxpayers may also need to understand how their transactions were classified, which rules were applied, why particular reliefs were included or excluded, and what assumptions influenced the final result.

One major challenge is the interpretation of financial records. Bank statements and other financial documents often contain short or ambiguous transaction descriptions. These descriptions may not clearly indicate whether a transaction represents employment income, business income, investment income, a transfer between accounts, or another financial activity. Automated text-processing techniques can help organise and retrieve information from textual descriptions, although their outputs still require appropriate validation [14]. The quality of transaction interpretation is important because incorrect classifications may affect subsequent tax-related processing.

Another challenge is applying the correct legislation for the relevant year of assessment. Tax rules can change through amendments, making it necessary to maintain accurate, year-specific information. Digital transformation initiatives in tax administration highlight the role of connected technologies and digital services in supporting tax-related activities [2], [3], [20]. However, digitalisation alone does not guarantee accurate calculations. The system must still apply the correct rules and provide sufficient information for users to review the results.

Personalisation also presents a challenge. Individuals have different income sources, financial commitments, eligibility conditions, and future goals. A tax strategy that may be suitable for one person may not be appropriate for another. Recommender systems provide established approaches for identifying and ranking options according to user characteristics and preferences [13]. In a tax advisory context, however, recommendations must also consider legal eligibility, financial circumstances, and potential consequences rather than relying solely on a ranking score.

Access to understandable tax information is another concern. Relevant requirements may be distributed across the principal Act and subsequent amendments, which can make them difficult for non-specialists to interpret. Transformer-based language models have expanded the possibilities for natural-language interaction and automated text understanding [8]. Nevertheless, language models may produce inaccurate or unsupported responses, particularly when dealing with complex or changing information [9]. Retrieval-Augmented Generation (RAG) addresses part of this challenge by combining language generation with retrieved information from external knowledge sources [7]. In a tax-related application, retrieved information must still be checked for relevance, currency, and legal accuracy.

Explainability and security are also important considerations. Users need to understand the basis of recommendations and calculations, while systems that handle financial information must protect sensitive data and reduce risks associated with unauthorised access. Model-explanation techniques such as SHAP can help explain the contribution of input features to individual predictions [12]. Responsible AI guidance emphasises the need to identify and manage risks throughout the development and use of AI systems [4]. Security practices, including those described in the OWASP API Security Top 10, are also relevant when designing applications that exchange information through APIs [19].

These challenges indicate the need for an integrated decision-support system that combines financial transaction interpretation, year-specific tax calculations, evidence-supported explanations, personalised recommendations, and the responsible handling of financial information.

1.1.3 Overview of the proposed AI-driven tax advisory system

The research project, titled An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka, aims to address

these challenges through an integrated application containing four complementary components. The system is intended to help users move from financial information to tax-related results and explanations that are easier to understand, inspect, and review.

Component A: Transaction Semantic Reasoning

This component focuses on interpreting financial transactions and organising them into structured categories that can support subsequent tax-related processing. Text-processing and information-retrieval approaches provide relevant foundations for working with transaction descriptions and other textual financial information [14]. However, transaction classifications should be treated carefully when the available description is incomplete or ambiguous.

Component B: Optimization and Explainable Engine

This component focuses on applying tax rules relevant to the selected year of assessment, carrying out deterministic tax calculations, and presenting explanations supported by relevant legal sources. Retrieval-based approaches can help connect answers and explanations to supporting documents [7]. The component's calculations must nevertheless remain grounded in the applicable tax rules rather than treating language-model outputs or machine-learning predictions as authoritative tax calculations.

Component C: Intelligent Tax Advisory Language Model

This component provides a natural-language interface through which users can ask tax-related questions and receive answers supported by relevant information. Transformer architectures have become an important foundation for modern language-processing systems [8], while RAG offers an approach for combining retrieved evidence with generated responses [7]. Because language models may produce unsupported or inaccurate content, the system needs suitable mechanisms for checking responses and directing users towards supporting legal information [9].

Component D: Personalized Recommendation and Predictive Impact Modeling Engine

This component focuses on identifying and ranking potentially suitable tax strategies according to a taxpayer's financial profile and estimating their possible financial effects. Recommender-system research provides a foundation for ranking relevant options [13]. Machine-learning techniques, including gradient-boosting methods, can support prediction and ranking tasks [10], [11], [17]. Explainability techniques can also help users understand which input factors influence a model's predictions [12].

The four components have different responsibilities but are intended to work together within one application. The integrated approach aims to provide more complete support than a tool that only calculates tax or provides general information.

This direction is consistent with wider developments in digital tax administration, where connected digital capabilities can support different stages of tax-related processes [2], [3].

An important design principle is to distinguish legally significant tax calculations from machine-learning predictions. The calculation engine applies defined rules and rates, while AI and machine-learning methods support tasks such as transaction interpretation, information retrieval, recommendation, and prediction. Predictive models may estimate likely outcomes, but these estimates should not be presented as guaranteed financial results. For example, Monte Carlo simulation can be used to explore a range of possible outcomes under specified assumptions rather than predicting one certain future [15]. Similarly, Random Forests and gradient-boosting methods provide established approaches to predictive modelling, but their outputs depend on the quality and suitability of the data used [16], [17].

The use of AI in a financial and regulatory context also requires attention to transparency, privacy, and security. Model explanations can help users inspect predictions [12], while AI risk-management guidance provides a broader framework for identifying and managing potential harms [4]. The system must also consider relevant personal-data protection requirements, including Sri Lanka's Personal Data Protection Act and its amendments [18].

The proposed system is designed for the Sri Lankan personal income tax context. It is intended to support users in understanding their information, reviewing tax-related results, and considering possible decisions. It is not intended to replace official tax determinations, professional tax advice, or human review where these are necessary.

1.2 Stakeholders and user needs

1.2.1 Primary stakeholders

Individual taxpayers are the main intended users of the proposed system. They may need help organising financial information, identifying relevant income categories, estimating tax liabilities, understanding tax terminology, and exploring strategies that could suit their circumstances. They may also want to understand why a particular result or recommendation has been presented.

The integrated application is intended to bring transaction analysis, tax calculation, personalised recommendations, and conversational guidance into one place. A shared interface can make these capabilities easier to access as part of a connected workflow. However, actual usability and usefulness must be established through appropriate evaluation rather than assumed from the design.

Tax professionals and auditors are another relevant stakeholder group. They may need to review how transactions have been classified, which rules and reliefs have been applied, and how a calculation or recommendation was produced. They also need to compare outputs with source documents and the applicable legislation before relying on them in professional work.

For these users, transparency and the ability to verify results are important. Research on interpretable machine learning highlights the importance of developing sound approaches to understanding model behaviour [5]. Techniques such as SHAP can provide explanations of the contribution of input features to model predictions [12]. However, an explanation of a prediction does not automatically establish that the prediction is correct. Outputs still need to be checked against appropriate evidence.

1.2.2 Secondary stakeholders

The Inland Revenue Department (IRD) is an important institutional reference point because the system operates within Sri Lanka's income tax framework. The legislation and amendments published by the IRD provide the principal legal sources against which the system's tax rules should be checked [1]. Digital tax administration research also provides context for considering how technology can support tax-related services [2], [3]. The proposed system may organise and explain tax information, but it does not replace the IRD's statutory responsibilities and is not an official tax assessment service.

Tax advisory firms and accounting or financial organisations are potential beneficiaries of tools that help structure financial information and support preliminary tax analysis. Such organisations may value consistent outputs, traceable calculations, and supporting information that can be reviewed before advice is given. Recommendation and predictive-modelling techniques may also help explore options based on individual financial circumstances [13]. Any professional use, however, would require suitable validation and review.

Researchers and students may also find the project useful as an example of how AI, machine learning, information retrieval, explainability, and predictive modelling can be applied in a regulated financial setting. The project brings together research areas such as RAG [7], ranking models [11], predictive modelling [16], and explainable AI [12]. It also raises practical questions about the reliability of synthetic data, the evaluation of model outputs, and the responsible use of AI in financial decision support.

These groups are identified as potential stakeholders and beneficiaries. Their inclusion in this report does not imply that all of them have formally tested the system or provided feedback.

1.2.3 Common user needs and system requirements

Although stakeholders may use the system for different reasons, several needs are shared across the groups.

First, users need financial records to be interpreted and organised into meaningful categories. Text-retrieval and text-processing methods can support the organisation of transaction descriptions [14], but uncertain classifications should remain open to review where necessary.

Second, users need tax calculations to follow the rules that apply to the selected year of assessment. The system should make the relevant assumptions and applied rules clear enough for users to check the result against official legislation [1]. Digital tax administration literature provides a broader context for the use of connected technology in tax-related processes [2], [3].

Third, users need recommendations that take account of their individual circumstances. Recommendation systems can help rank relevant options [13], while predictive models can estimate possible outcomes based on available data [10], [17]. Nevertheless, a recommendation should not be treated as suitable simply because it appears in a ranked list. Legal eligibility, affordability, data quality, and potential consequences must also be considered.

Fourth, users need clear answers to tax-related questions. RAG can help connect generated responses to retrieved information [7], but the quality of the answer depends on the quality and relevance of the retrieved sources. Since language models can produce inaccurate or unsupported statements [9], responses about legal requirements should provide supporting information where possible and should not remove the need for verification.

Fifth, users need to understand the basis of model-generated recommendations and predictions. Explainable machine-learning research addresses how model behaviour can be examined [5], and SHAP provides a method for explaining feature contributions to predictions [12]. Such explanations can support review, although they do not replace independent validation.

Finally, users need their financial information to be handled responsibly. Appropriate data protection and API security measures are important because the system may process sensitive personal and financial information [18], [19]. AI risk-management guidance also emphasises identifying, assessing, and managing risks throughout system development and use [4].

The four components address these needs in complementary ways. Component A supports transaction interpretation; Component B supports rule-based tax calculations and traceability; Component C supports access to understandable tax information; and Component D supports personalised strategy ranking and potential financial-impact analysis.

The intended outcome is a consistent user experience with clear boundaries between component responsibilities. Whether the integrated system meets user expectations, produces reliable results, and provides meaningful explanations must be assessed through testing and evaluation.

1.3 Alignment with the Sustainable Development Goals

The project has potential links to the United Nations Sustainable Development Goals (SDGs), particularly SDG 9, SDG 10, and SDG 16 [6]. These links arise from the

project's focus on digital innovation, access to understandable information, and transparency in tax-related decision support. They should be understood as intended contributions rather than proof that the project has already achieved measurable social or economic impact.

1.3.1 SDG 9: Industry, Innovation and Infrastructure

SDG 9 promotes resilient infrastructure, sustainable industrialisation, and innovation [6]. This project relates to the goal through the development of an integrated digital application that combines AI, machine learning, information retrieval, and rule-based processing to address a practical financial and regulatory problem.

The system brings together transaction classification, tax calculation, personalised recommendations, and tax-information retrieval. Developing these functions within one application provides an opportunity to investigate how different technical approaches can support connected stages of a decision-making process. This is relevant to the wider movement towards digital services in tax administration discussed in OECD publications [2], [3].

The project also investigates how different computational methods can serve different purposes. Retrieval-based methods can connect language-model responses with supporting documents [7], while predictive methods such as gradient boosting can support recommendation and modelling tasks [10], [17]. These methods should be selected and evaluated according to the task rather than treated as interchangeable solutions.

The project's contribution to SDG 9 is primarily its technical approach and investigation of an integrated decision-support application. Further evidence would be needed to demonstrate its effectiveness in practice, scalability, or suitability for wider adoption.

1.3.2 SDG 10: Reduced Inequalities

SDG 10 focuses on reducing inequalities within and among countries [6]. In the personal income tax context, some people may find it difficult to access or understand tax information, especially if they have limited tax knowledge or cannot easily obtain professional advice.

The proposed system aims to make information easier to navigate by bringing tax calculations, legal explanations, and personalised recommendations together. A conversational interface supported by retrieved information may help users access relevant explanations without needing to interpret every legal document independently [7]. Personalised recommendations may also help users explore options that reflect their financial circumstances [13].

Clearer information may help reduce some knowledge-related barriers. However, a digital system cannot by itself guarantee equal access or reduce inequality. The

benefits may depend on affordability, digital literacy, accessibility, language support, data quality, and the accuracy of the information provided.

For this reason, the project's relationship with SDG 10 should be described as a potential contribution to more accessible tax information. Demonstrating a measurable reduction in inequality would require further research into actual use, accessibility, and user outcomes.

1.3.3 SDG 16: Peace, Justice and Strong Institutions

SDG 16 promotes peaceful and inclusive societies, access to justice, and effective, accountable, and inclusive institutions [6]. For this project, the most relevant themes are access to reliable regulatory information, transparency, and accountability.

Tax-related outputs can be difficult to check when users cannot identify the rules or assumptions behind them. Without supporting information, users may struggle to understand why a result was produced or whether they should seek further professional advice.

The proposed system is designed to address this concern by using year-specific tax rules, deterministic calculations, supporting legal references, and explanations of outputs. RAG provides a technical approach for grounding generated answers in retrieved documents [7], while explainability methods can help users examine model predictions [12]. These approaches can support transparency, but they do not guarantee that all outputs will be correct. The relevance of retrieved sources, the accuracy of tax rules, and the quality of the validation process remain important.

Privacy and security are also relevant to responsible operation. Sri Lanka's personal data protection legislation provides a legal context for handling personal information [18], while API security guidance highlights risks that can arise when applications exchange data through interfaces [19]. These issues should be considered alongside the broader risk-management principles described in AI guidance [4].

The system is not a government service and does not replace statutory decisions, professional review, or authoritative legal interpretation. Its potential contribution to SDG 16 depends on the accuracy of the tax rules, the quality of supporting evidence, the effectiveness of validation, and the responsible handling of user information.

Overall, the project has its clearest direct connection to SDG 9 through digital innovation. Its focus on making tax information easier to understand and review also creates potential links to SDG 10 and SDG 16 [6]. These are intended contributions, and further evaluation would be required to establish measurable development outcomes.

1.4 Purpose and scope of the common report

1.4.1 Purpose of the common report

This common integrated solution report documents the overall problem, design, development, integration, and evaluation of the four-component tax advisory system developed by group R26-DS-004. Its purpose is to explain how the individual components contribute to one shared objective and how their combined functions address challenges in financial transaction interpretation, Sri Lankan income tax calculation, personalised recommendations, and tax-related communication.

The individual theses focus on the research and development of specific components. This report takes a system-level view. It describes the relationship between the components, the overall architecture, the integration process, and the requirements that apply to the complete application.

The report also discusses the approach to evaluation, the extent to which common objectives have been achieved, the limitations that remain, and the system's potential value. The technical approaches used across the system draw on established research in areas such as retrieval-augmented language generation [7], predictive modelling [10], [16], [17], recommendation and ranking [11], [13], and explainable AI [12]. These sources provide research context; they do not, by themselves, demonstrate the performance of the project's implementation.

1.4.2 Scope of the integrated solution

The integrated solution covers four main areas. The first is the interpretation and classification of financial transactions into information relevant to tax assessment. The second is the application of relevant Sri Lankan tax rules to calculate tax and provide traceable results. The third is the provision of understandable, evidence-supported answers to tax-related questions. The fourth is the ranking of personalised tax-strategy recommendations and estimation of their possible financial effects.

These functions are intended to be accessed through a shared application. The report therefore considers both the capabilities of each component and the way their inputs, outputs, and responsibilities relate to one another.

The report also considers cross-component matters such as data exchange, shared interfaces, data structures, error handling, reliability, usability, security, privacy, explainability, and responsible AI use. These matters are important when designing and evaluating trustworthy systems [4], [5]. Data protection and API security are also relevant because the application handles information through software interfaces and may process personal financial data [18], [19].

At a technical level, the project's wider research context includes information retrieval [14], language-model grounding [7], predictive modelling [10], [16], [17], recommendation ranking [11], [13], and simulation of uncertain future outcomes [15]. These references support the methodological context for the system. The exact

methods implemented, their evaluation procedures, and their measured results should be described in the relevant methodology and component-specific chapters.

The project focuses on Sri Lankan personal income tax. It does not cover every form of taxation and does not replace official tax administration processes. The system is presented as a decision-support tool to help users understand tax-related information and consider possible decisions.

1.4.3 Relationship between the common report and individual theses

The common report and the four individual theses have complementary roles. Each individual thesis provides detailed research and development evidence for its component, while this report explains how the four contributions are intended to work together as one solution.

The four components are summarised below.

Component A: Transaction Semantic Reasoning

This component focuses on interpreting and classifying financial transactions and preparing structured financial information for later tax-related processing. Text-processing and information-retrieval research provides relevant background for working with textual descriptions [14].

Component B: Optimization and Explainable Engine

This component focuses on applying tax rules relevant to the selected year of assessment, carrying out deterministic tax calculations, and presenting traceable explanations supported by legal sources. Retrieval-augmented methods provide a relevant approach for connecting information to supporting documents [7].

Component C: Intelligent Tax Advisory Language Model

This component focuses on retrieving and communicating tax-related knowledge through natural-language responses supported by relevant evidence. Transformer-based language models and RAG are relevant to this functionality [7], [8], while research on language-model limitations highlights the need for verification [9].

Component D: Personalized Recommendation and Predictive Impact Modeling Engine

This component focuses on ranking potentially suitable tax strategies, estimating adoption likelihood, explaining model outputs, and modelling possible financial impacts. Ranking and recommendation research [11], [13], predictive-modelling methods [10], [16], [17], explainability techniques [12], and Monte Carlo simulation [15] provide relevant methodological background for these tasks.

The components have separate responsibilities but are connected through the information they use and produce. Financial information interpreted by Component A can support subsequent tax processing. Component B provides the rule-based foundation for tax calculations. Component D uses relevant taxpayer and tax information to produce personalised recommendations, while Component C helps users understand tax-related information through natural-language interaction. The later architecture and integration chapters describe the relationships between these functions in more detail.

This common report is limited to the integrated solution and the evidence available from the group's research and development activities. It does not reproduce every literature review, algorithm, implementation detail, or experiment in full. Instead, it explains the collective contribution of the components to the overall research problem.

The report also distinguishes between intended functionality and demonstrated results. Statements about implementation, testing, performance, stakeholder feedback, and system effectiveness must be supported by actual project evidence. The system is presented as a decision-support solution for Sri Lankan personal income tax, not as a replacement for professional advice or an official tax determination.

The remaining chapters discuss the overall problem definition and component mapping, the integrated architecture, the integration and implementation process, system-level evaluation, and the project's potential commercial, sustainability, and responsible-innovation implications. The final chapter summarises the main integrated contribution and identifies directions for further research and development.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Personal income tax compliance in Sri Lanka requires individuals to interpret their financial information, identify taxable income, determine applicable reliefs, apply the correct tax rules for the relevant year of assessment, and understand the basis of their tax liabilities. These activities can become difficult when financial records contain unclear transaction descriptions, tax legislation changes over time, and relevant information is distributed across different legal documents and information sources. The Inland Revenue Act, No. 24 of 2017, and its subsequent amendments establish the legislative framework that must be considered when determining personal income tax obligations [1].

The central problem addressed by this research is the **difficulty of providing accurate, understandable, traceable, and personalised support for Sri Lankan**

personal income tax compliance through a single integrated process. Existing approaches may address individual aspects of this process, such as calculating a tax amount or providing general tax information, without necessarily connecting financial transaction interpretation, assessment-year-specific tax calculations, legal explanations, and personalised financial recommendations.

For example, a taxpayer may have access to financial records but still face difficulties determining which transactions are relevant to tax assessment. Even when income information has been identified, the taxpayer must establish which legal provisions apply to the relevant assessment year and understand how the resulting tax liability has been calculated. In addition, a calculation alone may not explain the legal reasoning behind the result or help the taxpayer explore strategies that are appropriate for their individual financial circumstances.

These difficulties are connected rather than independent. Incorrectly interpreted financial information may affect subsequent tax processing, while unclear explanations can make it difficult for users to verify the basis of a result. Similarly, recommendations may have limited practical value if they do not account for the taxpayer's circumstances or the applicable tax rules. Therefore, addressing only one part of the process may leave other important user needs unresolved.

The research consequently investigates how these related challenges can be addressed through an integrated decision-support system designed for Sri Lankan personal income tax. The common problem is not simply the absence of an artificial intelligence application; rather, it is the difficulty of connecting financial information, legally grounded calculations, understandable advisory information, and individualised recommendations in a coherent and verifiable workflow.

The proposed system is intended to assist users in understanding their tax-related circumstances and making more informed decisions. It is not intended to replace the statutory responsibilities of the Inland Revenue Department, authoritative interpretations of tax legislation, or professional tax advice. Its effectiveness must be assessed using appropriate technical and functional evidence.

2.2 Root causes and overall research gap

2.2.1 Fragmented financial information and transaction interpretation

One of the underlying causes of the common problem is the difficulty of interpreting financial information before it can be used for tax-related processing. Bank statements and similar records may contain abbreviations, short descriptions, repeated transaction patterns, or insufficient contextual information. A transaction description alone may not clearly establish the purpose of a payment or its tax treatment.

As a result, financial information may require classification and review before it can be used in subsequent calculations. If transaction categories are assigned incorrectly, the resulting tax analysis may be affected. This creates a need for a systematic way to

interpret transactions, organise relevant information, and allow uncertain classifications to be reviewed.

The research gap addressed by this part of the project concerns the preparation of structured, tax-relevant financial information from transaction records that may contain ambiguous descriptions. The Transaction Semantic Reasoning component addresses this subproblem by supporting transaction interpretation and classification. However, classification results must not automatically be treated as definitive legal decisions about taxability when additional context or human verification is required.

2.2.2 Changes in tax legislation and the need for traceable calculations

A second cause is the need to apply the correct tax rules for a particular year of assessment. Personal income tax calculations depend on the applicable legislation, rates, reliefs, thresholds, and other relevant provisions. Amendments to legislation mean that rules must be checked against the appropriate legal sources rather than assumed to remain unchanged across assessment years [1].

A calculation that produces a final tax amount without identifying its underlying rules may be difficult for a taxpayer or professional to verify. Users may need to understand which provisions were applied, why particular reliefs were considered, and how the final result was obtained. This is particularly important in a regulatory domain where calculations should be consistent with the applicable legal requirements.

The research gap addressed by this part of the project is the need to connect year-specific tax rules with deterministic calculations and explanations that can be traced to supporting legal sources. The Optimization and Explainable Engine addresses this requirement by applying defined tax rules and providing supporting information for the resulting calculations. Its reliability depends on the correctness, completeness, and appropriate assessment-year coverage of the underlying legal rules.

2.2.3 Difficulty accessing understandable and evidence-supported tax guidance

A third cause is the difficulty of understanding tax legislation and obtaining clear explanations of tax-related requirements. Legal documents may contain technical terminology and provisions that are difficult for individuals without a tax background to interpret. Users may need explanations of particular concepts, clarification of how a provision applies, or guidance on where relevant information can be found.

General-purpose conversational systems may make information easier to access, but their responses may not always provide sufficient evidence for claims about legal requirements. In a tax-related setting, a response should be grounded in relevant and reliable information, and users should be able to inspect the sources supporting important explanations.

The research gap addressed by this part of the project concerns the communication of tax-related information through understandable natural-language responses supported by retrieved evidence. The Intelligent Tax Advisory Language Model addresses this subproblem by providing a conversational interface for tax-related questions and explanations. Its outputs must still be checked against the underlying sources, particularly where legal interpretation or an individual tax decision is involved.

2.2.4 Limited personalisation and uncertainty in tax-strategy recommendations

A fourth cause is that tax-related recommendations may not be equally suitable for every individual. Taxpayers can differ in their income sources, financial commitments, available resources, eligibility for particular strategies, and long-term objectives. A strategy that may be appropriate for one individual may be unsuitable or financially impractical for another.

Presenting general recommendations without considering these differences may limit their usefulness. In addition, the immediate estimated benefit of a strategy may not fully represent its longer-term financial consequences. Users may need to compare possible options while considering factors such as affordability, suitability, risk, and uncertainty in future financial conditions.

The research gap addressed by this part of the project is the need to rank potentially suitable tax strategies according to individual financial circumstances and estimate their possible financial impact. The Personalized Recommendation and Predictive Impact Modeling Engine addresses this subproblem through personalised recommendation and predictive modelling. Such recommendations should be understood as decision-support outputs rather than guarantees of tax savings, adoption, or future financial performance.

2.2.5 Overall research gap

The four underlying challenges are closely related, but each requires a different type of processing. Transaction interpretation prepares financial information for further analysis. Rule-based tax computation applies the relevant legislation. Natural-language advisory capabilities help users understand tax-related information. Personalised recommendation and predictive modelling support the comparison of strategies based on individual circumstances.

The overall research gap is therefore the need to bring these functions together in a coherent system while maintaining clear responsibilities between them. A system that focuses only on transaction classification may not establish the correct tax liability. A calculator may produce a result without making it sufficiently understandable. A conversational interface may explain general concepts without calculating an individual's tax liability, while a recommendation engine may generate options that are not suitable unless the relevant financial and tax information has been considered.

This research addresses the integration of these complementary functions within a common decision-support application. It investigates how financial transaction interpretation, deterministic tax calculation, evidence-supported advisory responses, and personalised financial impact analysis can work together to support the Sri Lankan personal income tax process.

The research does not assume that combining these functions automatically guarantees accuracy or effectiveness. Instead, the integrated solution must be evaluated through component-level testing, integration testing, and appropriate end-to-end evaluation. This is necessary to determine whether the system performs its intended functions and whether information can be passed between components consistently and reliably.

2.3 Subproblem and component mapping

The overall problem is divided into four related subproblems. Each subproblem has a corresponding component with a defined responsibility. Although the components are distinct, they are interdependent because the overall decision-support process requires their capabilities to work together.

2.3.1 Subproblem A: Interpreting and classifying financial transactions

Financial records may contain transaction descriptions that are ambiguous, abbreviated, or insufficient to determine their meaning directly. Before such information can support tax-related processing, relevant transactions need to be interpreted and organised into appropriate categories.

The subproblem addressed by Component A is how to identify and classify relevant information from financial transactions while accounting for uncertainty in transaction descriptions. The component is intended to produce structured information that can be used by other parts of the integrated system.

Corresponding component: Component A - Transaction Semantic Reasoning.

Expected contribution: To organise financial transaction information into meaningful categories and support subsequent tax-related processing.

Evidence for evaluation: Transaction classification test cases, labelled evaluation data where available, confusion matrices, precision, recall, F1-score, and examples of correctly and incorrectly classified transactions. Where relevant, evaluation should also examine ambiguous cases and the need for manual review.

2.3.2 Subproblem B: Applying assessment-year-specific tax rules and explaining calculations

Tax calculations must follow the applicable legislation for the selected year of assessment. The challenge is to apply the relevant rules consistently and provide enough information for users to understand and verify the resulting calculations.

The subproblem addressed by Component B is how to represent and apply tax rules for supported assessment years while producing deterministic calculations and traceable explanations. The component must distinguish between rules supported by the underlying legal sources and information that requires further verification.

Corresponding component: Component B - Optimization and Explainable Engine.

Expected contribution: To calculate tax using applicable assessment-year-specific rules and provide explanations and legal references supporting the results.

Evidence for evaluation: Tax calculation test cases, comparison against independently verified expected results, rule-coverage checks, assessment-year validation, tests of relief eligibility and tax-rate application, and verification that explanations refer to the appropriate supporting provisions. Calculation accuracy should be reported using actual test results rather than assumed performance.

2.3.3 Subproblem C: Providing understandable and evidence-supported tax guidance

Taxpayers may struggle to understand legal terminology and locate relevant information across tax legislation and related documents. Although conversational interfaces can make information easier to access, their responses must remain grounded in appropriate sources.

The subproblem addressed by Component C is how to provide understandable natural-language responses to tax-related questions while retrieving and using relevant supporting information. The component should help users understand tax concepts and locate relevant information without presenting unsupported statements as authoritative legal advice.

Corresponding component: Component C - Intelligent Tax Advisory Language Model.

Expected contribution: To support conversational access to tax-related knowledge through responses grounded in relevant retrieved evidence.

Evidence for evaluation: A defined set of tax-related questions, relevance of retrieved sources, correctness of answers against verified reference material, citation or source-support checks, and tests for unsupported claims or incomplete answers. Where expert evaluation is available, reviewer assessments may also be reported.

2.3.4 Subproblem D: Generating personalised recommendations and estimating financial impact

Taxpayers have different financial circumstances and may not benefit equally from the same strategies. Recommendations therefore need to consider individual profiles, relevant eligibility conditions, financial feasibility, and possible future outcomes.

The subproblem addressed by Component D is how to identify and rank potentially suitable tax strategies for an individual and estimate their possible financial impact under defined assumptions. The component must distinguish estimated outcomes from guaranteed savings and communicate the uncertainty associated with future financial projections.

Corresponding component: Component D - Personalized Recommendation and Predictive Impact Modeling Engine.

Expected contribution: To rank potentially suitable strategies based on individual financial information and estimate possible financial consequences to support comparison and decision-making.

Evidence for evaluation: Recommendation-ranking measures where suitable, eligibility and feasibility checks, tests using defined taxpayer profiles, comparison of ranking outputs against expected or reference rankings, and validation of projected financial impacts against the assumptions used in the models. If adoption predictions are evaluated using synthetic or proxy labels, that limitation should be stated clearly. Where Monte Carlo simulation is used, the simulation assumptions and output distributions should also be documented.

2.3.5 Relationship between the four components

The four components address different stages of the overall tax-related decision-support process. Component A focuses on the interpretation of financial transactions. Component B applies the relevant tax rules and produces traceable calculations. Component C provides a natural-language interface for understanding tax-related information. Component D supports personalised strategy selection and the analysis of possible financial outcomes.

Their responsibilities are complementary rather than interchangeable. Transaction classification does not independently determine the legal tax treatment of every transaction. Tax calculation requires applicable rules and appropriate financial inputs. Conversational explanations require relevant supporting information, while personalised recommendations depend on suitable taxpayer information and defined assumptions.

The integrated solution is intended to connect these capabilities through a common application. The exact operational sequence and data exchanged between components depend on the implemented architecture and are described in the later

integration chapters. Each component should first be evaluated according to its own responsibilities, followed by integration testing to determine whether the complete workflow operates consistently.

2.4 Overall aim and group objectives

2.4.1 Overall aim

The overall aim of this research is to **design, develop, integrate, and evaluate an AI-driven decision-support system for Sri Lankan personal income tax that combines financial transaction interpretation, assessment-year-specific tax calculation, evidence-supported natural-language advisory, and personalised tax-strategy recommendation with predictive impact modelling**.

The aim focuses on the coordinated operation of the four components and their ability to support users in understanding tax-related information and exploring possible decisions. It also recognises the importance of traceability, appropriate validation, and clear separation between deterministic tax calculations and machine-learning-based interpretation or prediction.

2.4.2 Group objectives

The following objectives translate the overall aim into activities that can be evaluated using technical and functional evidence from the project.

Objective 1: Establish an integrated solution architecture.

Design and implement a common application structure that connects the four components and defines their responsibilities, inputs, outputs, and communication requirements.

Evaluation evidence: Architecture documentation, component interface definitions, API or integration test results where applicable, and evidence that the components can exchange the required information.

Objective 2: Support the interpretation of financial transactions for tax-related processing.

Develop and evaluate a transaction interpretation and classification capability that organises relevant financial records into meaningful categories for subsequent processing.

Evaluation evidence: Transaction classification test cases, labelled evaluation data where available, classification metrics, and documented examples of ambiguous or incorrectly classified transactions.

Objective 3: Support tax calculations using applicable assessment-year-specific rules.

Implement and evaluate deterministic tax calculations based on the rules available for the assessment years supported by the system, with explanations and references that allow the results to be reviewed.

Evaluation evidence: Independently verified calculation examples, rule-coverage checks, assessment-year tests, and verification of the supporting explanations and legal references.

Objective 4: Provide understandable tax-related responses supported by evidence.

Develop and evaluate a conversational advisory capability that retrieves relevant tax-related information and uses it to support responses to user questions.

Evaluation evidence: A test set of tax-related questions, retrieval relevance checks, answer correctness assessments, source-support verification, and documented examples of unsupported or incomplete responses.

Objective 5: Generate personalised tax-strategy recommendations and estimate potential financial impact.

Develop and evaluate a recommendation capability that considers individual financial circumstances, ranks potentially suitable strategies, and estimates possible outcomes under defined assumptions.

Evaluation evidence: Recommendation-ranking tests, eligibility and feasibility checks, taxpayer-profile test cases, predictive evaluation results where appropriate, and documentation of financial projection assumptions and limitations.

Objective 6: Evaluate the integration and end-to-end behaviour of the complete system.

Assess whether the four components operate together as intended and whether information can move through the integrated application without unacceptable inconsistencies or processing failures.

Evaluation evidence: Integration test cases, end-to-end scenarios, API or service-level test results where applicable, error-handling tests, and documented observations of failures and corrective actions.

Objective 7: Examine the limitations and responsible use of the integrated solution.

Identify limitations affecting accuracy, reliability, interpretability, privacy, usability, and the practical application of the system in the Sri Lankan personal income tax context.

Evaluation evidence: Test findings, documented limitations, review of data-handling practices, analysis of model assumptions, and stakeholder feedback where such feedback has actually been collected.

2.4.3 Mapping of objectives to components and evaluation evidence

Table 2.1 summarises the relationship between the common objectives, the components responsible for addressing them, and the evidence required to evaluate their achievement.

Table 2.1: Mapping of group objectives to components and evaluation evidence

Group objective	Main component(s)	Evaluation evidence
Establish an integrated architecture	Components A, B, C, and D	Architecture documentation, interface checks, and integration tests
Interpret financial transactions	Component A	Classification test cases and appropriate classification metrics
Apply assessment-year-specific tax rules	Component B	Verified calculation cases, rule coverage, and legal-reference checks
Provide evidence-supported tax guidance	Component C	Retrieval relevance, answer correctness, and source-support checks
Generate personalised recommendations and impact estimates	Component D	Ranking evaluation, feasibility checks, and financial projection validation
Evaluate end-to-end integration	All four components	Integrated scenarios, interface tests, and error-handling results
Examine responsible use and limitations	All four components	Documented limitations, validation evidence, data-handling review, and available stakeholder feedback

The table identifies the evidence that should be used to evaluate each objective. It does not imply that every evaluation has already been completed or that a particular performance level has been achieved. The final assessment of each objective should be based on the actual results documented in the implementation and evaluation chapters and the four individual theses.

2.5 Scope assumptions and constraints

2.5.1 Scope of the integrated solution

The project focuses on personal income tax in Sri Lanka. Its intended scope includes the interpretation and classification of relevant financial transactions, tax calculation using supported assessment-year-specific rules, natural-language access to tax-

related information, and personalised recommendations with estimates of possible financial impact.

The common report concentrates on the overall architecture, relationships between the components, integration process, shared requirements, and evaluation of the complete solution. Detailed algorithms, component-specific implementation decisions, and individual experimental results are addressed in the corresponding component theses.

The scope is limited to the functionality actually implemented and supported by the group's technical evidence. It should not be assumed that the system supports every possible income category, tax relief, assessment year, taxpayer profile, or legal scenario. The exact supported coverage must be documented in the relevant technical and evaluation sections.

2.5.2 Assumptions

The development and evaluation of the system depend on several assumptions.

First, financial transaction information supplied for processing is assumed to be available in a format that the implemented system can accept. The usefulness of transaction classification depends on the quality, completeness, and context of the input data. Ambiguous transactions may require clarification or human review.

Second, tax calculations depend on the availability of appropriate legal rules for the assessment year being processed. Those rules must be checked against relevant legal sources, and the system should not treat unsupported or unverified rules as authoritative. If a required provision is unavailable or the assessment year is outside the implemented coverage, the limitation should be made clear.

Third, the advisory component depends on the availability and relevance of supporting tax-related information. A response should not be considered reliable merely because it is expressed fluently. Its content and supporting evidence must be evaluated.

Fourth, personalised recommendations and financial projections depend on the quality of the taxpayer information, the assumptions used by the models, and the strategy options supported by the system. Predicted adoption or financial outcomes should be treated as estimates rather than guarantees.

Finally, the system is assumed to operate as a decision-support application in which users may review important outputs and seek professional assistance when necessary. The system is not intended to replace official tax authorities or qualified tax professionals.

2.5.3 Technical and operational constraints

The development of the integrated solution is subject to technical and operational constraints. These may include the availability and quality of financial data, the supported formats of input records, the coverage of the tax-rule knowledge base, the performance of the individual models, and the reliability of communication between components.

The system's ability to provide correct tax-related outputs depends on the correctness and completeness of the implemented rules and the information supplied to it. Changes to tax legislation may require the relevant rules, supporting documents, and assessment-year configurations to be reviewed and updated. Therefore, the system's outputs should be interpreted in relation to the legal coverage and update status documented by the project.

The use of machine-learning models also introduces limitations. Transaction classification may be affected by ambiguous descriptions or data that differ from the examples used during model development. Recommendation rankings and predictive estimates may be affected by the representativeness of the data, the assumptions used for training, and the evaluation methodology. Where synthetic data or proxy labels are used, results may not fully represent real taxpayer behaviour.

The integration of four components also creates dependencies between services and data structures. Incompatible inputs, missing information, processing failures, or differences in assumptions can affect the end-to-end workflow. These risks require appropriate interface validation, error handling, and integrated testing.

2.5.4 Exclusions and limitations

The project is not intended to provide an official tax assessment, replace professional tax advice, or make statutory decisions on behalf of the Inland Revenue Department. Its purpose is to support users by organising financial information, applying implemented tax rules, presenting relevant explanations, and helping users explore potential strategies.

The system's recommendations should not be interpreted as guaranteed tax savings or as confirmation that a particular strategy is legally or financially appropriate for every individual. Users may need additional information or professional advice before acting on a recommendation.

Similarly, natural-language responses should not be treated as authoritative legal interpretations without checking the supporting sources. Any unsupported, incomplete, or uncertain response should be recognised as a limitation rather than accepted solely because it appears convincing.

The project also does not claim to demonstrate improvements in national tax compliance, reductions in inequality, or measurable economic benefits merely

through the development of the application. Such outcomes would require broader evaluation using suitable evidence beyond the implementation of the technical system.

Finally, the scope of the common report does not include every technical detail or experiment associated with the four components. Those details are documented in the individual theses, while this report concentrates on the integrated solution and its collective contribution.

2.5.5 Dependencies and conditions for successful evaluation

The successful operation and evaluation of the integrated solution depend on several conditions. The transaction interpretation component must produce structured information in a format that subsequent processing can use. The tax calculation component must apply the rules associated with the selected assessment year. The advisory component must retrieve relevant information to support its responses, and the recommendation component must receive suitable financial and tax-related inputs for generating personalised outputs.

These dependencies mean that component-level performance alone is not sufficient to establish the effectiveness of the complete system. Integration and end-to-end tests are also needed to determine whether information is passed correctly between components and whether the combined workflow produces consistent results.

The final evaluation should therefore distinguish between component-level achievements, successful integration, and broader claims about practical effectiveness. Any reported performance values, stakeholder feedback, or evidence of user acceptance must come from the actual tests and evaluation activities conducted during the project.

By defining these assumptions, constraints, exclusions, and dependencies, the project establishes a clear boundary for assessing the integrated solution. This provides a basis for the following chapters, which describe the architecture, component relationships, implementation, testing, and evaluation of the proposed system.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The proposed solution, titled *An Intelligent AI-Driven Decision Support System for Personalized Income Tax Compliance and Advisory in Sri Lanka*, is designed as an integrated application that brings together four complementary components to support the personal income tax process. Rather than treating transaction analysis, tax calculation, tax advisory, and financial recommendations as separate activities, the system aims to connect them within a common framework. This allows the different components to contribute to a shared objective: helping users understand their tax-related circumstances and make more informed decisions.

The need for an integrated approach arises because tax-related decision-making involves several connected tasks. Financial transactions must first be interpreted, applicable tax rules must be identified, calculations need to be performed correctly, and users may require explanations of the results. In addition, individuals may wish to explore tax strategies that suit their financial circumstances. Addressing these needs through isolated tools can make it difficult to maintain consistency between financial information, calculations, explanations, and recommendations.

The proposed system addresses this challenge through four components, each with a clearly defined responsibility.

Component A: Transaction Semantic Reasoning focuses on interpreting financial transactions and organising them into meaningful categories. Its output provides structured financial information that can support subsequent tax-related processing.

Component B: Optimization and Explainable Engine focuses on applying the relevant tax rules for supported assessment years. It performs deterministic tax calculations and provides explanations and supporting legal references to help users understand how results are obtained.

Component C: Intelligent Tax Advisory Language Model provides a conversational way for users to ask tax-related questions and obtain understandable responses grounded in relevant information. Its purpose is to make tax knowledge easier to access without relying solely on unsupported model-generated answers.

Component D: Personalized Recommendation and Predictive Impact Modeling Engine focuses on identifying and ranking potentially suitable tax strategies based on an individual's financial circumstances. It also estimates possible financial outcomes to help users compare alternatives.

The value of the integrated solution lies in the way these capabilities complement one another. Transaction interpretation helps organise financial information, the tax engine establishes the calculation results, the advisory component helps users understand tax-related concepts, and the recommendation component supports the exploration of individualised strategies. Together, they aim to provide a more complete decision-support experience than a calculator or a general conversational assistant operating alone.

An important principle of the architecture is the separation of deterministic tax calculations from machine-learning-based predictions. Tax calculations should follow the applicable rules and defined calculation logic, while artificial intelligence and machine-learning techniques support tasks such as transaction interpretation, information retrieval, recommendation ranking, and predictive analysis. This separation helps maintain clear responsibilities and makes it easier to identify which part of the system produced a particular result.

The solution is intended to be accessed through a common application interface, allowing users to interact with the system without needing to understand the internal operation of every component. The components must exchange information through defined interfaces and use compatible data structures to support the integrated workflow.

Overall, the system is intended to make tax-related information more accessible, calculations more traceable, and recommendations more relevant to individual circumstances. Its practical value depends on the accuracy of the implemented tax rules, the quality of the input data, the reliability of the individual components, and the results of integrated testing. The system therefore functions as a decision-support tool rather than a replacement for professional tax advice or official tax determinations.

3.2 High level architecture

3.2.1 Architectural overview

The high-level architecture of the proposed system follows a component-based approach in which four specialised components contribute to a shared application. A common frontend provides the user-facing interface, while a shared API layer supports communication between the frontend and the relevant backend services.

The frontend is intended to provide a consistent experience for users accessing the system's different capabilities. Through this interface, users can submit financial information, request tax calculations, ask tax-related questions, and explore personalised recommendations, depending on the functions available in the implemented application.

The shared API layer acts as the communication boundary between the user interface and the backend functionality. It receives requests from the frontend and directs them to the relevant component or service. The exact routing, authentication, and service-management responsibilities should be documented according to the implementation.

At the backend level, the four components perform their respective tasks. Component A interprets financial transaction information. Component B applies supported tax rules and generates calculation results with explanations. Component C retrieves relevant information and generates natural-language advisory responses. Component D evaluates individual financial information and generates personalised recommendations with estimated financial impacts.

The architecture also depends on external information sources, particularly financial records supplied by users and official tax legislation used to establish and verify tax rules. These sources have different roles: financial records provide information about the taxpayer's circumstances, while legislation provides the legal basis for tax-related calculations and explanations.

The components are connected through the common application, but their internal methods remain distinct. This design supports separation of responsibilities while allowing the complete system to deliver its intended functionality.

3.2.2 High-level architecture diagram

Figure 1 presents a conceptual view of the integrated architecture. It shows the user-facing application, the shared API layer, the four backend components, and the principal information sources. The diagram represents the logical structure of the solution; it should be checked against the actual implementation before being presented as the final deployed architecture.

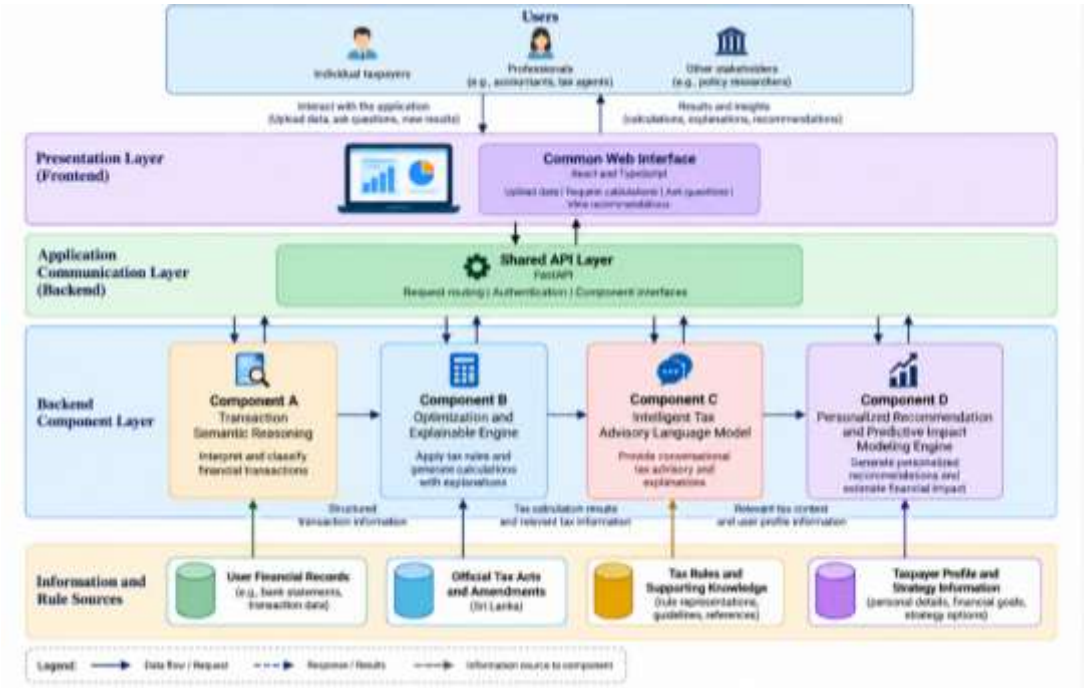


Figure 1: High-level architecture of the integrated AI-driven tax advisory system

In this architecture, the common web interface is the main point of interaction for users. The shared API layer connects user requests to the relevant backend component and returns the resulting information to the interface. The diagram also illustrates the principal logical dependencies between financial transaction processing, tax calculation, advisory information, and personalised recommendations.

Official tax legislation and its amendments provide the basis for verifying the rules used by the system. Financial records and taxpayer-profile information provide the individual data required for relevant processing. The architecture should preserve the distinction between information derived from user records, information established by tax rules, and predictions produced by machine-learning models.

The diagram is a logical representation rather than a complete deployment diagram. Specific details such as database placement, authentication services, cloud

infrastructure, network boundaries, and the deployment of individual services should be added only where they are confirmed by the project's actual configuration.

3.2.3 Main architectural layers

The architecture can be understood through three main layers.

Presentation layer

The presentation layer provides the common user interface through which users access the system. The project uses React and TypeScript with Vite for the frontend development environment. The interface is intended to make the system's functions accessible through a consistent interaction model and to present results in a form that users can understand.

Application communication layer

The application communication layer connects the frontend with the backend components. The project uses a FastAPI backend, with the shared API structure supporting communication between the application and its specialised functions. The API layer should define the expected request formats, response structures, and error-handling behaviour needed for consistent integration.

Backend component layer

The backend component layer contains the four specialised components. Each component addresses a particular part of the overall problem while contributing to the integrated workflow. Component A interprets transactions, Component B applies tax rules, Component C provides natural-language advisory responses, and Component D generates personalised recommendations and predictive impact estimates.

The components may also depend on supporting information, including financial records, tax legislation, tax-rule representations, and taxpayer-profile information. The precise storage arrangements and connections to these sources should be described in the implementation chapter according to the technologies actually used.

3.2.4 Architectural design principles

The architecture follows several principles that support the integration of the four components.

First, **separation of responsibilities** ensures that each component focuses on a defined task. This reduces unnecessary overlap and makes individual components easier to test and maintain.

Second, **consistent interfaces** help the components exchange information in predictable formats. Clear request and response structures are important for avoiding integration errors and maintaining compatibility as components are updated.

Third, **traceability** supports the review of tax calculations and advisory responses. Where applicable, outputs should retain the information needed to identify the rules, sources, or assumptions used to produce them.

Fourth, **appropriate use of artificial intelligence** distinguishes between tasks that benefit from learned models and calculations that should follow explicit rules. Machine-learning outputs should not replace the underlying legal rules used to determine tax liabilities.

Finally, **modularity and maintainability** allow individual components to be improved without requiring the entire application to be redesigned. However, changes to one component must still be checked for their effect on shared interfaces and end-to-end behaviour.

These principles provide the architectural foundation for the integration strategy and implementation activities discussed in later chapters.

3.3 Component 1 summary

3.3.1 Component overview

Component name: Transaction Semantic Reasoning

Responsible component researcher: Component A researcher (insert the group member's name in the final report).

Component A addresses the difficulty of interpreting financial transaction descriptions and organising financial records into meaningful categories for tax-related processing. Financial statements may contain short descriptions, abbreviations, and incomplete contextual information, making it difficult to determine the meaning of individual transactions from the descriptions alone.

The component focuses on transforming transaction information into a more structured representation that can support the other parts of the integrated system. Its role is to interpret and classify financial information rather than independently determine the final legal tax treatment of every transaction.

3.3.2 Inputs and core approach

The principal inputs are financial transaction records, including transaction descriptions and other available transaction attributes supported by the implementation. The quality of the classification depends on the information available in these records and the ability of the component to interpret the descriptions.

The component uses its implemented transaction interpretation and classification methods to organise transactions into relevant categories. The exact model architecture, preprocessing steps, training data, classification labels, and evaluation procedures are documented in the individual Component A thesis.

Because a transaction description may not always provide enough information to establish its purpose, ambiguous or uncertain cases may require additional context or human review. This is important because a predicted transaction category should not automatically be treated as a definitive legal conclusion about taxability.

3.3.3 Outputs and contribution to the integrated system

The expected output is structured transaction information containing the assigned categories and any additional fields supported by the implementation. This information can then be used by subsequent tax-related processing, subject to appropriate validation.

Component A contributes to the integrated solution by reducing the need to interpret every transaction manually and by preparing financial information in a more consistent form. It provides the transaction-level foundation for further analysis while preserving the distinction between transaction classification and formal tax determination.

Its effectiveness should be evaluated using suitable classification test data, relevant performance metrics, and examples of ambiguous or incorrectly classified transactions. Detailed methods and results remain the responsibility of the individual Component A thesis.

3.4 Component 2 summary

3.4.1 Component overview

Component name: Optimization and Explainable Engine

Responsible component researcher: Component B researcher (insert the group member's name in the final report).

Component B addresses the challenge of applying the correct tax rules for a selected year of assessment and explaining how tax calculations are produced. Since tax legislation can change over time, the calculation process must use the relevant rules rather than assume that the same provisions apply to every assessment year.

The component provides the rule-based calculation foundation of the integrated system. Its primary responsibility is to apply defined tax rules and produce calculation results that can be checked against the underlying legal provisions.

3.4.2 Inputs and core approach

The component receives the financial and taxpayer information required by its supported calculation functions, together with the relevant assessment year. Where transaction information is supplied by Component A, that information must be suitable for the calculation process and should be checked when necessary.

The engine applies deterministic calculation logic using the tax rules available for the supported assessment years. Supporting rules and legal information are used to establish the applicable rates, reliefs, thresholds, and other conditions required by the implemented calculations. The project also uses YAML files to represent tax rules, while supporting information and the precise storage structure should be documented according to the implementation.

The calculation process should make a clear distinction between verified legal rules, assumptions, and information that is not supported by the available rule set. If a required rule or assessment year is not supported, the system should not silently substitute an unrelated rule.

The detailed rule representation, extraction process, validation logic, calculation algorithms, and source-verification mechanisms are documented in the individual Component B thesis.

3.4.3 Outputs and contribution to the integrated system

The expected outputs include calculated tax results and supporting explanations or legal references, where these are available in the implementation. These outputs provide a basis for users to understand the calculation and review the provisions used.

Component B contributes to the overall system by establishing a consistent, rule-based approach to tax calculation. It helps separate legally grounded calculations from predictions produced by machine-learning models and provides relevant tax information that can support advisory explanations and personalised recommendations.

The component should be evaluated using independently verified calculation examples, assessment-year tests, checks of applicable rules and reliefs, and verification of supporting explanations. Detailed results should be reported only after the corresponding tests have been performed.

3.5 Component 3 summary

3.5.1 Component overview

Component name: Intelligent Tax Advisory Language Model

Responsible component researcher: Component C researcher (insert the group member's name in the final report).

Component C addresses the difficulty of understanding tax legislation and obtaining clear answers to tax-related questions. Although official legal documents provide the necessary information, their terminology and structure can make them difficult for users without a tax background to interpret.

The component provides a conversational interface intended to help users access tax-related knowledge in a more understandable form. Its responsibility is to communicate relevant information and explanations rather than independently replace the deterministic tax calculation process.

3.5.2 Inputs and core approach

The component receives tax-related questions or requests for explanations from users. It uses the advisory methods implemented in the project to retrieve relevant information and generate natural-language responses grounded in supporting material.

The quality of an answer depends on the relevance and completeness of the retrieved information, the accuracy of the generated response, and the ability to connect important claims to appropriate sources. Where the available information does not adequately support an answer, the system should recognise that limitation rather than present an unsupported response as authoritative.

The exact language model, retrieval approach, knowledge representation, source-grounding mechanisms, and validation procedures are described in the individual Component C thesis. These details should match the implemented system rather than being inferred from the overall architecture.

3.5.3 Outputs and contribution to the integrated system

The expected outputs are natural-language responses that explain tax-related concepts, clarify relevant requirements, or direct users to supporting information. Where the implementation supports source references, these should help users inspect the basis of the response.

Component C contributes to the integrated system by making tax-related information easier to access and understand. It complements the calculation engine by helping users interpret relevant information and complements the recommendation engine by supporting users who need to understand tax concepts before comparing potential strategies.

The component should be evaluated using a defined set of tax-related questions, checks of retrieved-source relevance, assessments of answer correctness, and verification that important statements are supported by the available evidence. Any

limitations in retrieval coverage or answer reliability should be reported in the individual thesis.

3.6 Component 4 summary

3.6.1 Component overview

Component name: Personalized Recommendation and Predictive Impact Modeling Engine

Responsible component researcher: Component D researcher (insert the group member's name in the final report).

Component D addresses the difficulty of identifying tax-related strategies that are appropriate for individual financial circumstances. Different taxpayers may have different income levels, financial commitments, eligibility conditions, and long-term objectives. As a result, the same strategy may not be equally suitable for every person.

The component focuses on ranking potentially suitable strategies and estimating their possible financial consequences. It extends the system beyond calculating an existing tax liability by helping users compare alternatives under defined assumptions.

3.6.2 Inputs and core approach

The component uses taxpayer-profile information, relevant financial features, available strategy information, and applicable eligibility conditions supported by the implementation. Where relevant, information produced by other components can provide the context needed to assess the suitability of a recommendation.

The project work for Component D includes recommendation ranking, adoption-likelihood estimation, and predictive impact modelling. The individual thesis describes the implemented techniques, including ranking models, adoption prediction, explainability methods, and simulation of potential multi-year outcomes where applicable.

Recommendations depend on the quality of the supplied profile information, the available strategy catalogue, the assumptions used by the models, and the suitability of the underlying evaluation data. Where synthetic data or proxy labels are used, the results must be interpreted in light of those limitations.

Financial projections are estimates rather than guarantees. Their usefulness depends on the assumptions used for variables such as future income, inflation, investment returns, and other factors included in the implemented model.

3.6.3 Outputs and contribution to the integrated system

The expected outputs include ranked recommendations and estimates of their possible financial impact, together with supporting explanations where these are available. The intention is to help users compare alternatives rather than automatically select or execute a strategy on their behalf.

Component D contributes to the overall solution by making the advisory process more individualised. While Component B focuses on tax calculations and Component C supports natural-language explanations, Component D considers which available strategies may be suitable for a particular financial profile and how their outcomes may differ under defined assumptions.

Its evaluation should examine recommendation ranking, eligibility and feasibility checks, predictive performance where appropriate, and the assumptions underlying financial projections. If the system estimates adoption likelihood using proxy labels, this should be stated explicitly, as predicted adoption is not equivalent to observed taxpayer behaviour.

3.7 End to end user and data flow

3.7.1 User entry and information submission

The end-to-end process begins when a user accesses the common application and selects the functionality required. Depending on the implemented workflow, the user may provide financial transaction records, taxpayer-profile information, a tax calculation request, or a question about tax-related matters.

The common interface sends the relevant request through the shared API layer to the appropriate backend component. The information must be provided in a format supported by the receiving component, and validation should be performed before further processing.

At this stage, the system needs to distinguish between financial information supplied by the user, information retrieved from legal or knowledge sources, and information generated by the system. Keeping these categories distinct helps avoid confusion when results are presented later.

3.7.2 Financial transaction interpretation

When the user submits financial transaction records for analysis, Component A interprets the available transaction information and assigns categories according to its implemented classification approach. The output is a structured representation of the financial information that can support subsequent tax-related processing.

Where a transaction is unclear or the available information is insufficient, the classification may require review or additional context. The system should not

assume that every transaction can be classified with certainty from its description alone.

The structured output from Component A can then be used by the tax calculation workflow, provided it meets the input requirements of the relevant component. This creates the first major connection between the components: financial transaction interpretation prepares information for further analysis without independently establishing the final tax liability.

3.7.3 Tax rule application and calculation

When a tax calculation is required, Component B receives the relevant financial and taxpayer information together with the selected year of assessment. It identifies and applies the rules supported for that assessment year and performs the calculation using deterministic logic.

The calculation produces a result together with supporting information, such as the rules or reliefs applied and relevant legal references, where available. These explanations help users or professionals inspect the calculation and identify information that may need further verification.

The tax calculation component provides the rule-based foundation for the integrated workflow. Its output can support the user's understanding of their tax-related circumstances and provide relevant information for subsequent advisory or recommendation activities. Any recommendation that depends on the calculation must use the applicable outputs and assumptions rather than independently inventing a tax liability.

3.7.4 Tax-related questions and explanations

Users may also ask questions about tax concepts, applicable requirements, or information presented by the system. Component C processes these requests through the implemented advisory workflow, retrieving relevant information and generating a natural-language response.

Where a question relates to a particular tax rule or calculation, the advisory component should use the relevant supporting information available to it. It should not treat a fluent response as a substitute for checking the underlying legal source or the deterministic calculation result.

The response is returned through the shared application interface, allowing the user to obtain explanations without needing to interpret every legal document independently. If the available evidence is incomplete or does not adequately support an answer, the limitations should be made clear.

This component complements the tax calculation engine by focusing on communication and explanation. The calculation engine establishes results according

to its implemented rules, while the advisory component helps users understand tax-related information in natural language.

3.7.5 Personalised recommendations and predictive impact analysis

When a user wishes to explore potential strategies, Component D uses the relevant taxpayer-profile information and the strategy options supported by the system. Where applicable, tax-related information and calculation outputs from Component B can provide additional context for evaluating potential options.

The recommendation engine ranks potentially suitable strategies according to its implemented methods and estimates possible financial outcomes under defined assumptions. The resulting information can help users compare alternatives based on factors such as predicted impact, suitability, affordability, and other criteria supported by the model.

The results should be presented as decision-support information rather than guaranteed savings or predictions of actual future behaviour. Users may need to verify eligibility, provide additional information, or seek professional advice before acting on a recommendation.

Component D therefore builds on the broader tax-related information available in the system and adds an individualised perspective to the decision-making process.

3.7.6 Results presentation and user review

After a component completes its task, its output is returned through the shared API layer to the common interface. Depending on the request, the user may receive structured transaction classifications, a tax calculation with explanations, a natural-language advisory response, or a ranked list of recommendations with estimated financial impacts.

The interface should present these outputs in a way that helps users distinguish between calculated results, retrieved legal information, machine-generated classifications, and predictive estimates. This distinction is important because these outputs have different levels of certainty and different purposes.

Users should be able to review important results and identify when additional information or professional verification may be necessary. The system supports the decision-making process but does not automatically make official tax determinations or guarantee the suitability of a recommendation.

3.7.7 Component interaction and exception handling

The integrated workflow depends on the components exchanging information in compatible formats. A classification result from Component A must meet the input requirements of the subsequent process. Component B must receive sufficient

information to calculate tax for a supported assessment year. Component C must have relevant supporting information for evidence-based responses, while Component D requires suitable profile information and strategy data to generate meaningful recommendations.

If required information is missing, a year of assessment is unsupported, a service fails, or the available evidence is insufficient, the system should handle the situation explicitly. Appropriate handling may include validation messages, requests for additional information, clear indications that a function is unavailable, or a recommendation for human review, depending on the implemented behaviour.

The specific error-handling mechanisms should be documented and tested as part of the integration process. It is important not to assume that the complete workflow will always succeed simply because each component operates individually.

3.7.8 Overall end-to-end contribution

The end-to-end workflow illustrates how the four components contribute to a common tax-related decision-support process. Component A organises financial transaction information, Component B applies the relevant tax rules, Component C helps users understand tax-related information, and Component D supports the comparison of personalised strategies and their possible financial outcomes.

The components are designed to complement one another while retaining distinct responsibilities. This arrangement allows the integrated system to address several connected user needs within one application rather than requiring users to rely on separate tools for transaction interpretation, calculation, explanation, and recommendation.

The overall effectiveness of the workflow depends on accurate inputs, appropriate legal-rule coverage, reliable component interfaces, relevant supporting information, and suitable evaluation. Consequently, the integrated system must be assessed not only through individual component tests but also through end-to-end scenarios that demonstrate how information moves between components and how the final outputs are presented to users.

4. COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

4.1.1 Overview of the Integration Approach

The proposed AI-driven tax advisory system consists of four interconnected components designed to address different aspects of personal income tax compliance and advisory in Sri Lanka. Although each component has a distinct responsibility, their combined operation provides users with a more comprehensive tax advisory experience. The integration approach therefore focuses on establishing clear communication between components, maintaining consistency in the information exchanged, and ensuring that the output of one component can support the activities of another.

The four components are:

- **Component A: Transaction Semantic Reasoning** – interprets and classifies financial transactions.
- **Component B: Optimization and Explainable Engine** – applies relevant tax rules, performs tax calculations, and provides explanations.
- **Component C: Intelligent Tax Advisory Language Model** – supports users by providing understandable, evidence-supported tax guidance.
- **Component D: Personalized Recommendation and Predictive Impact Modeling Engine** – generates personalised recommendations and estimates their potential financial impact.

The system follows a layered architectural approach consisting of a presentation layer, an application communication layer, a backend component layer, and an information and rule source layer. The common web interface provides the primary point of interaction, while the shared API layer acts as the communication point between the interface and the backend components. This structure separates user interaction from component-specific processing and supports the coordinated operation of the overall system.

4.1.2 Integration Sequence and Component Dependencies

The integration process is organised around the functional dependencies between the four components. Component A processes financial transaction information and produces structured outputs that can be used by Component B. Component B applies the relevant tax rules to the available information and produces calculation results and explanations. These results, together with relevant taxpayer profile information,

can support the personalised analysis performed by Component D. Component C provides a complementary advisory function by using relevant tax knowledge to respond to users' questions.

The principal relationships between the components are summarised below.

Integration relationship	Purpose
Component A to Component B	Supplies structured transaction information for tax assessment.
Component B to Component D	Provides tax calculation results and relevant tax information for recommendation analysis.
Tax knowledge sources to Component C	Supports the generation of evidence-based tax guidance.
Taxpayer profile to Component D	Provides personal and financial context for recommendations.
Shared API layer to all four components	Supports communication between the web interface and the component services.
Component outputs to the web interface	Makes classifications, calculations, explanations, advisory responses, and recommendations available to users.

Table 2: Principal Integration Relationships Between the Four System Components

These relationships describe the intended functional dependencies of the integrated architecture. The precise execution order may vary depending on the feature selected by the user. For example, a user asking a tax-related question may interact with Component C without first executing the complete transaction-classification and recommendation workflow.

4.1.3 Integration Environments and Configuration

The system uses a React and TypeScript-based web interface and a FastAPI backend. These technologies provide the foundation for user interaction and backend communication. The frontend presents the available functions, collects user inputs, and displays the results returned by the backend. The backend provides the communication layer through which requests can be directed to the appropriate component.

During development and integration, the frontend, shared API layer, and individual components should be configured to use consistent request formats, service endpoints, and data representations. Configuration differences between development, testing, and deployment environments should also be considered when validating the integrated system.

Environment-specific details, including service URLs, dependency versions, configuration files, and deployment settings, should be documented according to the actual implementation. Sensitive values, such as API keys and credentials, should not be included in source code or exposed through the user interface.

4.1.4 Integration Responsibilities and Risk Management

Successful integration requires each component to maintain a clearly defined responsibility while following the agreed communication requirements. Component developers are responsible for explaining their component's inputs, outputs, dependencies, and known limitations. The integration process then brings these functions together through the shared application architecture.

Potential integration risks include inconsistent data formats, missing input values, differences in tax-rule representations, incompatible component outputs, and failures during communication between services. These risks can be reduced through interface documentation, input validation, consistent data models, component-level testing, and end-to-end testing.

Particular attention is required when combining financial transaction information with tax calculations and personalised recommendations. An incorrect transaction classification or incomplete taxpayer profile may affect subsequent outputs. Therefore, the integrated system should identify incomplete or invalid inputs and communicate relevant limitations rather than silently treating uncertain information as correct.

4.2 Interfaces, Data Contracts, and Shared Models

4.2.1 Shared API Interface and Communication Responsibilities

The shared API layer provides a common communication point between the React and TypeScript frontend and the backend components. Its main responsibility is to receive application requests, direct them to the relevant functionality, and return the resulting information to the user interface.

This separation helps prevent the frontend from becoming tightly coupled to the internal implementation of every component. Instead, the interface can interact with the backend through defined operations, while each component remains responsible for its own processing logic.

The exact API routes, HTTP methods, request schemas, and response structures should be documented from the implemented FastAPI application. These details should be included in the final report only after they have been verified against the actual source code or API documentation.

4.2.2 Data Exchange Between the Four Components

The integrated system depends on the consistent exchange of information between components. Each component must receive the information required for its task and produce outputs that can be interpreted by the receiving component.

The principal data exchanges are described below.

a. Financial transaction information

Component A receives financial transaction records and interprets their meaning for subsequent tax-related processing. Its outputs may include transaction classifications and associated information needed by the tax calculation process. The structure and fields of these outputs should follow the actual transaction model implemented in the system.

b. Tax calculation information

Component B uses relevant financial information and applicable tax rules to calculate tax-related results. Its outputs may include calculated amounts, relevant rule references, and explanations of the calculation. Where Component D uses these results, the exchanged information must retain sufficient context to avoid confusing values calculated under different assumptions or assessment years.

c. Tax advisory information

Component C uses relevant tax-related knowledge to generate responses to user questions. The information supplied to this component should be sufficient to support the user's question and, where applicable, provide traceable references to the relevant tax provisions or supporting material.

d. Taxpayer profile and recommendation information

Component D uses taxpayer-related information and relevant tax results to produce personalised recommendations and estimate potential financial impacts. Its input data should distinguish between information provided by the user, calculated values, and assumptions used for predictive analysis.

4.2.3 Shared Data Models and Validation Rules

Shared data models provide a consistent representation of information exchanged throughout the system. Relevant data may include financial transactions, assessment-year information, tax calculation results, taxpayer profile details, advisory responses, and recommendation outputs.

To support reliable communication, data exchanged between components should be validated for completeness, type correctness, and logical consistency. For example, monetary values should use a consistent representation, assessment-year identifiers should follow an agreed format, and required fields should be checked before processing begins.

Validation should also account for missing or ambiguous information. If a transaction does not contain sufficient details for reliable classification, the system should avoid presenting an uncertain classification as definitive. Similarly, tax calculations should not proceed on the assumption that missing information

represents a valid zero value unless that interpretation is explicitly justified by the applicable business rules.

Where applicable, exchanged records should preserve information about the source of the data, the relevant assessment year, and the assumptions used in calculations or predictions. These details help maintain consistency across components and make the results easier to review.

4.2.4 Error Responses and Interface Compatibility

The integration layer should support consistent handling of invalid requests, missing information, unavailable component services, and unexpected processing errors. Errors should be communicated in a format that allows the frontend to present an understandable message without exposing sensitive implementation details.

For example, an invalid request may require the user to correct the submitted information, while a temporary service failure may require the user to retry the operation. Where a component cannot produce a reliable result, the system should communicate the limitation rather than generate an unsupported output.

Changes to shared data models and API interfaces should also be managed carefully. If a component changes the structure of an output, the receiving components and frontend must be checked for compatibility before the updated interface is used in the integrated system. Versioning should be documented where it is implemented; otherwise, it should be treated as a recommended improvement rather than an existing feature.

4.3 Shared Non-Functional Requirements

4.3.1 Performance and Responsiveness

The integrated system should process user requests within a reasonable period and present results without unnecessary delay. Performance depends on several factors, including the complexity of financial transaction processing, tax calculations, retrieval of supporting knowledge, and predictive impact analysis.

The shared architecture should avoid unnecessary repeated processing and should provide clear feedback when an operation takes longer than expected. Performance evaluation should consider both individual component execution times and the overall time required to complete representative user workflows. Actual response times should be reported only when they have been measured during testing.

4.3.2 Usability and Accessibility

The system is intended to assist users who may have different levels of financial and tax-related knowledge. Therefore, the interface should present technical information in clear language and organise the results so that users can understand the purpose of each function.

Tax calculations should be accompanied by meaningful explanations, while advisory responses should distinguish relevant information from assumptions or limitations. Recommendations should explain their intended purpose and the factors that influence their estimated impact.

The interface should also use consistent terminology, readable text, clear navigation, and understandable validation messages. Accessibility considerations include keyboard navigation, sufficient colour contrast, and meaningful labels for interactive elements. These requirements should be verified through appropriate usability and accessibility checks rather than assumed to have been fully achieved.

4.3.3 Security and Privacy of Financial Information

The system processes potentially sensitive financial and personal information. Appropriate security and privacy measures are therefore essential throughout data submission, processing, storage, and presentation.

The implementation should apply access controls appropriate to the available functions, validate incoming requests, and protect credentials and other sensitive configuration values. Personal and financial data should only be collected and processed when necessary for the relevant functionality. Data should not be exposed to unrelated users or included unnecessarily in application logs.

Where external language models or other third-party services are used, the handling of submitted information should be assessed against the system's privacy requirements and the relevant service configuration. The final report should identify the protections actually implemented and distinguish them from proposed future improvements.

4.3.4 Accuracy, Explainability, and Traceability

Accuracy is particularly important because incorrect tax information may influence a user's financial decisions or compliance activities. The system should therefore distinguish between deterministic tax calculations, transaction classifications, generated advisory responses, and predictive estimates.

Component B should apply the relevant assessment-year-specific tax rules and provide explanations that allow users to understand how a result was obtained. Component C should ground its responses in relevant supporting knowledge where the implementation supports this capability. Component D should communicate that its projected financial impacts depend on assumptions and may differ from actual future outcomes.

Where practical, results should retain references to their underlying information sources, calculation assumptions, or applicable tax provisions. The use of explainable outputs supports review and accountability, but it does not independently

guarantee that every result is correct. The relevant functions must still be tested against suitable reference cases.

4.3.5 Reliability and Maintainability

The system should remain understandable and manageable as individual components evolve. Separating the components by responsibility supports maintainability because changes to one function can be assessed without automatically requiring a redesign of the entire application.

Reliability also depends on validating inputs, handling exceptions, identifying failed operations, and avoiding misleading results when a required service or information source is unavailable. Error messages should help users understand whether they need to correct their input or try again later.

Maintainability can be supported through modular source code, clear interface documentation, consistent naming conventions, version control, and repeatable testing procedures. These practices should be documented according to the development process actually followed by the research team.

4.3.6 Scalability and Responsible Use

The architecture should allow the system to accommodate additional users, financial records, tax-related questions, and future enhancements without requiring all components to be redesigned simultaneously. However, the ability to scale depends on the deployment infrastructure, processing requirements, and actual workload. Scalability should therefore be treated as a design consideration until supported by load-testing evidence.

The system should also communicate the limitations of AI-generated guidance and predictive outputs. It is intended to support tax-related understanding and decision-making rather than remove the need for professional judgement in complex or uncertain cases. Users should be able to review the information underlying a result and recognise when additional verification is necessary.

These requirements are consistent with responsible AI principles that emphasise reliability, transparency, accountability, and the management of system risks [4], [5].

4.4 Integrated Workflow and Exception Handling

4.4.1 Normal End-to-End Workflow

The integrated workflow begins when a user accesses the common web interface and selects a tax-related function. Depending on the selected task, the system directs the request to the relevant backend component through the shared API layer.

For a workflow involving financial transaction analysis and personalised recommendations, the main stages are as follows:

1. **User input and submission:** The user provides the relevant financial records, assessment-year information, or taxpayer profile details required by the selected function.
2. **Input validation:** The system checks whether the submitted information is in an acceptable format and contains the required fields.
3. **Transaction interpretation:** Component A interprets and classifies financial transactions where transaction processing is required.
4. **Tax calculation and explanation:** Component B applies the relevant tax rules to the available information and produces tax-related results and explanations.
5. **Advisory support:** Component C responds to tax-related questions using relevant supporting knowledge. This function may operate independently or alongside other system functions, depending on the user's request.
6. **Personalised recommendation:** Component D uses relevant taxpayer information and tax-related results to generate recommendations and estimate potential financial impacts.
7. **Presentation of results:** The frontend displays the outputs returned by the relevant components, allowing the user to review the available information and explanations.

This sequence represents the principal integrated workflow rather than a requirement for every user request to execute all four components. The actual execution path depends on the functionality selected and the dependencies implemented in the application.

4.4.2 Handling Invalid or Incomplete User Input

The system may receive incomplete transaction records, incorrectly formatted values, missing assessment-year information, or insufficient taxpayer details. Such cases should be identified before the affected processing operation proceeds.

Where the missing information can reasonably be supplied by the user, the interface should explain what needs to be corrected. Where the information is optional, the system should apply a default only if that default is explicitly defined and appropriate for the relevant function.

If the available information is insufficient to produce a reliable result, the system should communicate that limitation. This is particularly important when incomplete inputs could affect tax calculations or personalised recommendations.

4.4.3 Handling Component and Communication Failures

An integrated system may encounter failures when a component cannot process a request, an API operation returns an error, or a required information source is unavailable. The system should distinguish these cases from valid results and communicate the problem in a way that supports recovery.

For example, a failed transaction-classification request should not be treated as if all transactions had been classified successfully. Likewise, a failed tax calculation should not be presented as a valid zero tax liability, and an unavailable recommendation function should not be represented as evidence that no suitable recommendations exist.

Where retrying an operation is appropriate, the system may allow the user to submit the request again. Repeated attempts should be controlled to avoid unnecessary processing or duplicate operations. If a component failure prevents a dependent function from operating, the system should identify the dependency and avoid generating a misleading downstream result.

The precise retry policies, logging mechanisms, timeouts, and fallback behaviour should be documented according to the implemented system. Where these mechanisms have not been implemented or tested, they should be identified as areas for improvement.

4.4.4 Handling Uncertain Tax Information and AI-Generated Outputs

Tax-related information may be uncertain because of incomplete user input, ambiguity in a financial transaction, differences between assessment years, or limitations in the available supporting knowledge. The system should preserve these distinctions when presenting its results.

For tax calculations, the applicable assessment year and relevant assumptions should be clear. For advisory responses, supporting references should be provided where available, and unsupported statements should not be presented as confirmed legal requirements. For predictive recommendations, estimated impacts should be distinguished from guaranteed financial outcomes.

If the system cannot identify sufficiently reliable supporting information, it should communicate that the result requires further verification instead of presenting an unsupported answer with unwarranted confidence.

4.4.5 Recovery, Result Consistency, and User Feedback

Following an error, the system should allow the user to understand the nature of the problem and determine the next appropriate action. Depending on the situation, this may involve correcting submitted information, retrying a request, reviewing an explanation, or seeking further professional guidance.

The outputs presented to the user should also remain consistent with the information used to generate them. For example, a recommendation should not rely on a tax calculation associated with a different assessment year without explicitly accounting for that difference.

User feedback and testing observations can help identify recurring integration problems and opportunities for improvement. However, feedback collection, automated recovery, and monitoring capabilities should be described as implemented features only when they are supported by the actual application.

4.5 Team Contribution and Dependency Matrix

4.5.1 Distribution of Responsibilities Across the Research Team

The project is organised around four individual research components that contribute to a common AI-driven tax advisory system. Each component addresses a distinct research problem while contributing to the shared objectives of improving transaction interpretation, tax calculation, tax-related guidance, and personalised decision support.

The contribution of each team member should be documented in a manner that distinguishes individual research responsibilities from collaborative activities. Individual responsibilities may include problem analysis, literature review, design, implementation, testing, evaluation, and documentation. Shared responsibilities may include architectural design, interface coordination, integration testing, and preparation of the common thesis report.

The final report should identify each student by their correct name and map their contribution to the relevant component and shared activities. The matrix below provides a structure for recording these responsibilities.

4.5.2 Component Ownership and Individual Contributions

Component	Research and development responsibility	Main dependency	Individual evidence to include
Component A: Transaction Semantic Reasoning	Design and development of financial transaction interpretation and classification	Financial transaction records and the interface used to exchange structured outputs	Individual thesis, implementation evidence, classification test cases, and evaluation results
Component B: Optimization and Explainable Engine	Design and development of tax calculation, rule application, and calculation explanations	Relevant financial information, assessment-year-specific tax rules, and supporting legal sources	Individual thesis, tax-rule implementation evidence, calculation test cases, and evaluation results
Component C:	Design and	Relevant tax	Individual thesis,

Intelligent Tax Advisory Language Model	development of tax-related question answering and advisory responses	legislation, supporting knowledge, and the user request	advisory implementation evidence, representative question-answer tests, and evaluation results
Component D: Personalized Recommendation and Predictive Impact Modeling Engine	Design and development of personalised recommendations and predictive financial impact analysis	Taxpayer profile information, relevant tax results, and prediction assumptions	Individual thesis, recommendation implementation evidence, predictive evaluation, and explanation of limitations

Table 3: Component Ownership and Individual Contributions

The evidence listed in this table represents the material that should be collected to demonstrate each contribution. It does not imply that every item has already been completed or that any particular performance result has been achieved.

4.5.3 Shared Integration and Testing Responsibilities

In addition to component-specific work, the research team must coordinate the interfaces and shared functionality required for the integrated application. These activities include agreeing on the data exchanged between components, checking the compatibility of component outputs, integrating the frontend with the backend, and evaluating representative end-to-end workflows.

The following matrix can be completed using the team's actual development and testing records.

Shared activity	Main responsibility	Evidence to record
Overall system architecture	Research team collaboratively	Architecture diagram, design decisions, and interface documentation
Shared API and data contracts	Designated backend or integration owner, with input from component developers	API definitions, request and response examples, and validation rules
Frontend and backend integration	Designated integration owner and relevant component developers	Screenshots, request-response evidence, and integrated workflow tests
Component-level testing	Respective component researcher	Test cases, actual outputs, expected outputs, and recorded results
End-to-end testing	Research team	Test scenarios covering the

	collaboratively	relevant component dependencies and complete user workflows
Evaluation and limitations	Respective component researchers, consolidated by the team	Evaluation metrics, test evidence, identified limitations, and supporting analysis
Common thesis documentation	Research team collaboratively	Reviewed chapters, consistent terminology, figures, tables, and references

Table 4:Shared Team Activities, Responsibilities, and Supporting Evidence

The terms “designated owner” and “research team collaboratively” should be replaced or supplemented with the actual names and responsibilities agreed upon by the group. The matrix should reflect the work performed rather than assigning responsibility retrospectively without supporting evidence.

4.5.4 Component Dependencies and Coordination

The four components are connected through functional dependencies. Component A contributes structured transaction information to the tax calculation process, while Component B provides relevant tax-related results that may support Component D's recommendation and predictive analysis. Component C contributes a separate advisory capability based on relevant tax knowledge. The shared interface and API layer connect these functions to the overall application.

These dependencies require coordination throughout development. Changes to transaction classifications may affect the information available for tax calculation, while changes to tax-rule representations or calculation outputs may affect downstream recommendation logic. Similarly, changes to shared API contracts may require corresponding updates to the frontend and the affected backend components.

The team should therefore record important interface changes, communicate dependencies between researchers, and verify that integrated functionality remains consistent after changes are introduced. Where integration issues arise, the team should identify the affected interface or component, record the observed behaviour, and document the corrective action supported by testing evidence.

4.5.5 Evidence of Individual and Collective Achievement

The final assessment of team contributions should be supported by verifiable evidence. Individual contributions can be demonstrated through component source code, research documentation, design artefacts, test records, and evaluation results. Collective contributions can be demonstrated through the integrated application, shared architecture, cross-component test cases, and the common thesis report.

Version-control history, issue records, meeting notes, and documented integration decisions may provide additional evidence of how the work was coordinated, where

these records are available. Screenshots and demonstrations should correspond to the actual implementation and should not replace the underlying test results where quantitative evaluation is required.

By separating individual contributions from shared integration activities, the report can demonstrate both the independent research undertaken within each component and the collective work required to produce a coherent AI-driven tax advisory system.

5. INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

5.1.1 Overview of the Technology Stack

The integrated AI-driven tax advisory system combines web development technologies, backend services, structured tax rules, and AI-based processing to support personal income tax compliance and advisory in Sri Lanka. The common technology stack provides the foundation for connecting the four research components through a shared application architecture.

The principal technologies identified in the project include React, TypeScript, Vite, and FastAPI. In addition, structured tax-rule files and supporting tax information are used by the relevant components to perform their respective functions.

Technology or resource	Purpose within the system
React	Develops the interactive web interface through which users access the system's functionality.
TypeScript	Supports structured frontend development and type checking.
Vite	Provides the frontend development and build environment.
FastAPI	Provides the backend API interface for handling application requests and connecting them to relevant processing functions.
YAML files	Represent structured tax-rule information used by the relevant tax-processing functionality.
Financial and taxpayer information	Provides the information required for transaction interpretation, tax calculations, and personalised analysis.
Tax legislation and supporting knowledge	Provides the legal and informational basis for tax calculations and advisory responses.
Git and GitHub	Support source-code version control and repository collaboration, as evidenced by the team's development environment.

Table 5: Common Technology Stack of the Integrated Tax Advisory System

The selection of these technologies supports the separation of user interaction, backend processing, and component-specific responsibilities. This structure allows the research components to contribute to a common application while retaining their individual purposes.

5.1.2 Frontend Technologies and User Interaction

The frontend is developed using React and TypeScript, with Vite supporting the development and build process. It provides the main point of interaction between users and the integrated tax advisory system. Through the interface, users can access the available functionality, submit relevant information, and review the outputs returned by the backend.

React supports the development of reusable interface elements, while TypeScript helps maintain consistency in the structure of frontend data and interactions. Vite provides the tooling required to develop and build the application.

The frontend communicates with the backend through the application's API interface. This separation allows the user interface to present transaction classifications, tax calculations, advisory responses, and personalised recommendations without needing to contain all the underlying processing logic.

5.1.3 Backend Services and Component Processing

FastAPI is used as the backend framework for the application. It provides a structure for defining API operations, receiving requests, validating input data, and returning responses to the frontend. The backend communication layer connects user requests with the relevant component functionality.

The four components have distinct responsibilities. Component A interprets and classifies financial transactions, Component B applies tax rules and produces calculation explanations, Component C provides tax-related advisory responses, and Component D generates personalised recommendations and predictive impact estimates.

The backend coordinates these functions and supports the exchange of information required by dependent components. For example, structured transaction information may be used during tax calculation, while relevant calculation results and taxpayer profile information may support personalised recommendations.

The internal organisation of the backend should be described according to the actual source code. In particular, the report should distinguish between components implemented as separate services and components implemented as modules within a common backend application.

5.1.4 Tax Rules and Supporting Information Resources

Tax legislation and assessment-year-specific rules are essential information resources for the system. The relevant tax-processing functionality uses structured tax-rule information to support calculations, while the advisory component requires supporting knowledge to respond to users' questions.

YAML files are used in the project to represent structured tax-rule information. This format provides a readable way to organise rule definitions and maintain them separately from application logic. Nevertheless, each rule must be checked against the relevant legal provision and assessment year before it is relied upon for tax calculations.

Financial transactions, taxpayer profiles, calculation results, and supporting tax knowledge must also be represented consistently to meet the requirements of the components that use them. The final implementation description should identify any databases, retrieval mechanisms, external services, or additional storage technologies that are actually used.

5.1.5 Development Infrastructure and Version Control

The development environment includes Visual Studio Code, where the project source code and Git-related information can be viewed. The team's screenshot of the development environment shows the project workspace, a Git Graph with commit and merge history, branch information, and a Source Control panel displaying pending changes.

Git provides a mechanism for recording source-code changes and maintaining the history of development. GitHub can provide a shared repository for storing committed code and supporting collaboration, where the local repository is connected to the corresponding remote repository.

These facilities are useful for a multi-component research project because changes to one component may affect shared interfaces or dependent functionality. Version-control history can help the team review previous changes and understand how the integrated application has evolved.

The specific hosting environment, database infrastructure, and production deployment platform should be documented separately using verified project records.

5.2 Development and Integration Process

5.2.1 Component-Based Development Approach

The development process is organised around four research components that address different but related aspects of the overall tax advisory problem. This approach allows each component to focus on a defined responsibility while contributing to the objectives of the integrated system.

Component A focuses on financial transaction interpretation, Component B addresses tax calculations and explainability, Component C provides tax-related advisory functionality, and Component D focuses on personalised recommendations and predictive impact modelling.

Although the components have distinct purposes, their development must account for their dependencies. For example, the information produced by transaction classification may be required for tax calculations, while the results of those calculations may contribute to personalised recommendations.

The overall development process therefore requires component-level implementation alongside coordination of the shared interfaces and workflows. Individual component functionality must be checked before it is relied upon by dependent components.

5.2.2 Interface Design and Component Coordination

Interface coordination is important because the four components must exchange information in a consistent format. Each component needs to define the information required for its operation, the output it produces, and the way that output can be consumed by other components.

Component A must provide transaction information in a form that the relevant tax-processing functionality can interpret. Component B must return calculation results that can be presented to users and, where required, used by Component D. Component C must return advisory responses in a format suitable for the common interface.

Defining and reviewing these expectations helps reduce integration problems caused by inconsistent data structures or incompatible outputs. The shared API layer provides the communication point between the frontend and the relevant backend functionality.

The exact API routes, request formats, response structures, and validation rules should be documented from the implemented application. Any interface design practices that were not formally followed should not be presented as completed development activities.

5.2.3 Implementation and Integration of the Four Components

The integration of the four components involves connecting their individual processing functions to the common application architecture.

Component A: Transaction Semantic Reasoning

Component A interprets and classifies financial transactions. Its integration role is to provide structured information that can support subsequent tax-related processing.

The consistency of these outputs is important because downstream functions may rely on the resulting transaction classifications.

Component B: Optimization and Explainable Engine

Component B applies the relevant tax rules and produces tax calculations and explanations. Its integration role includes receiving the necessary financial information, applying the appropriate assessment-year-specific rules, and returning results that can be presented to users or consumed by dependent functionality.

Component C: Intelligent Tax Advisory Language Model

Component C supports users by providing tax-related advisory responses. Its integration role includes receiving user questions, using the relevant supporting knowledge through the implemented mechanisms, and returning responses suitable for presentation through the common interface.

Component D: Personalized Recommendation and Predictive Impact Modeling Engine

Component D generates personalised recommendations and estimates their potential financial impact. Its integration role includes using relevant taxpayer information, tax-related results, and predictive assumptions to produce recommendations that can be reviewed by users.

The integration of these components enables the application to provide several related functions through a common interface. The actual operation of each integration point should be demonstrated through implementation records and suitable testing evidence.

5.2.4 Version Control and Change Management

Git is used in the project's development environment to track source-code changes and maintain development history. The Visual Studio Code screenshot captured for this project shows the Source Control panel, a Git Graph containing commit and merge records, branch information, and files awaiting a commit.

This evidence demonstrates the presence of Git version-control activity and the state of the local working tree at the time the screenshot was taken. The commit graph can be used to review recorded development changes, while the Source Control panel identifies modifications that have not yet been committed.

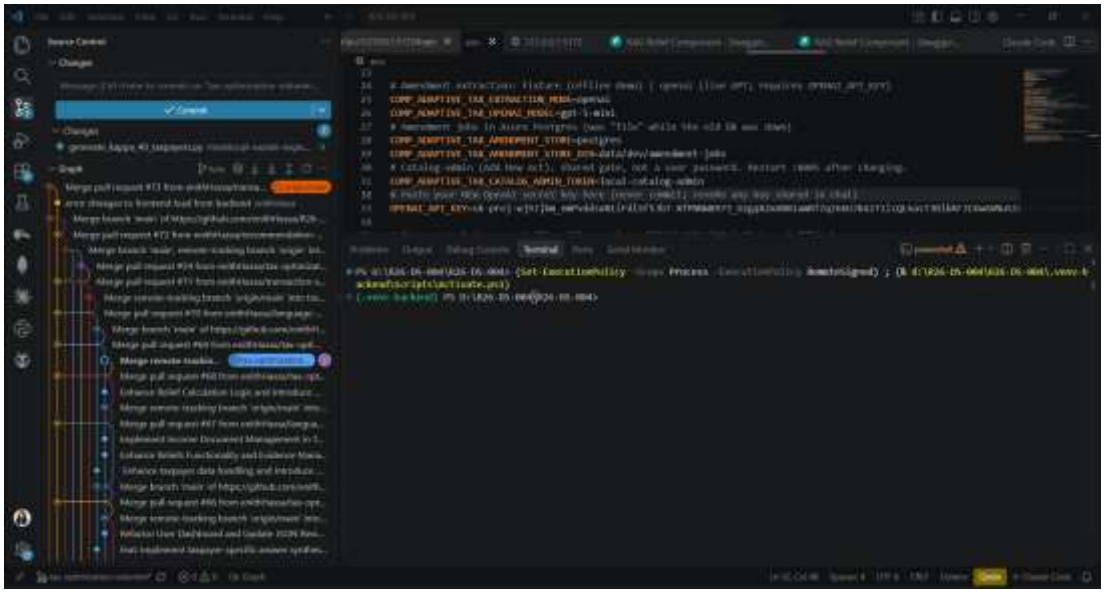


Figure 2: Git version-control history and working-tree status of the integrated tax advisory project.

Version control supports the management of changes across the frontend, backend, and component-specific functionality. Changes to shared interfaces should be reviewed carefully because modifications may affect dependent components or previously working workflows.

The available screenshot does not independently establish the use of formal code reviews, automated quality checks, or continuous integration. These practices should be described only if they are supported by the team's actual repository and development records.

5.2.5 Issue Management and Integration Testing

Development and integration may involve problems such as incompatible data structures, missing inputs, incorrect assumptions, or unexpected component behaviour. Recording these problems helps the team investigate their causes and document the corrective actions taken.

Relevant issue records may include the problem description, the affected component or interface, the expected behaviour, the observed behaviour, and the result of subsequent testing. Where the team used an issue tracker or project management tool, appropriate records can provide evidence of how development problems were managed.

Testing should cover individual component functionality, communication between components, and representative end-to-end workflows. Component-level tests examine individual functions, interface tests verify information exchange, and end-to-end tests evaluate the combined behaviour of the application.

The final report should distinguish between tests that were planned, tests that were executed, and tests that passed their defined acceptance criteria. Actual testing results and quantitative performance measures should be presented in Chapter 6.

5.3 Deployment and Operational Configuration

5.3.1 Application Build and Configuration

Preparing the application for use requires the frontend, backend, component dependencies, and supporting resources to be configured appropriately. The frontend can be prepared for deployment through the Vite build process, while the backend must expose the required API operations and access the resources needed by the components.

Configuration may include service addresses, dependency versions, application settings, and the locations of tax-rule files or other supporting resources. Environment-specific settings should be managed carefully to avoid exposing sensitive values in source code or publicly accessible frontend files.

The final report should include the actual build commands, startup procedures, and configuration settings used by the team. These details should be verified against the project's configuration rather than copied from generic deployment instructions.

5.3.2 Deployment Environment and Service Availability

The frontend and backend must be accessible through the environment in which the application is executed. The frontend must be able to communicate with the relevant backend endpoints, and the backend must be able to access the component functionality and supporting resources required to process requests.

The deployment description should identify where the frontend is hosted, how the backend is started, and how supporting resources are made available. If the application is running locally during development, the report should describe it as a development environment rather than a production deployment.

The actual hosting provider, server configuration, and production deployment method should be added once confirmed from the team's records. A screenshot showing an application running locally demonstrates execution in that environment, but it does not by itself establish public production availability.

5.3.3 Operational Configuration and Security

Operational configuration helps maintain consistent behaviour across the frontend, backend, and component dependencies. Sensitive configuration values, including API keys and credentials, must be protected and should not be embedded in publicly accessible source code or included in screenshots used for documentation.

The application should validate incoming information and communicate appropriate errors when required inputs or resources are unavailable. If external AI services are used, their configuration and data-handling implications should also be considered.

The final report should distinguish between the security controls that were actually implemented and those recommended for future development. Evidence of a configuration file alone does not establish that the application has adequate security or access control.

5.3.4 Monitoring, Logging, and Error Diagnosis

Monitoring and logging can help developers identify failures and investigate unexpected application behaviour. Relevant information may include failed API requests, component-processing errors, missing configuration files, or problems accessing supporting resources.

Logs should provide sufficient diagnostic information without unnecessarily exposing personal taxpayer information or other sensitive data. User-facing error messages should also avoid exposing internal implementation details.

If the project includes monitoring or logging facilities, their operation should be documented using actual implementation and testing evidence. If the application was evaluated only in a local development environment without a dedicated monitoring system, that limitation should be stated clearly.

5.3.5 Backup, Recovery, and Maintenance

Backup and recovery procedures are important when an application depends on persistent data, configuration files, or maintained tax-rule resources. Appropriate backups can help restore required information following accidental deletion, configuration errors, or infrastructure failures.

Maintenance activities may include updating tax-rule information, correcting defects, updating dependencies, and checking that changes do not adversely affect existing functionality. Tax-rule updates require particular care because incorrect changes may affect calculations for a specific assessment year.

The final report should describe backup frequency, storage locations, restoration procedures, and recovery tests only if these have been established and verified. If formal backup and restoration procedures were outside the scope of the research prototype, this should be acknowledged as a limitation and considered for future development.

5.4 Evidence of Team Execution

5.4.1 Purpose of Development Evidence

Evidence of team execution demonstrates how the research team organised its work, coordinated component development, managed implementation activities, and progressed towards the integrated system. It complements the technical description by providing verifiable information about the work undertaken.

The common report should summarise the most relevant evidence rather than reproduce extensive raw activity logs. Evidence should be selected according to the claim it supports. For example, Git history can demonstrate recorded development activity, while integration test records can demonstrate whether particular workflows were evaluated.

The report should distinguish between evidence of activity and evidence of achievement. A commit demonstrates that a change was recorded, but it does not automatically prove that the corresponding functionality is correct or that the integrated system meets its evaluation objectives.

5.4.2 Project Planning and Task Coordination

Project planning records can demonstrate how the team divided responsibilities, tracked development activities, and coordinated work across the four components. If Microsoft Planner or another project management tool was used, relevant task assignments and milestones can be summarised in this section.

The evidence should focus on activities related to component development, interface coordination, integration testing, and common report preparation. It should also identify important dependencies or issues that affected progress, where supported by actual records.

If the team did not use Microsoft Planner, an equivalent task board or another genuine project planning record may be used instead. The report should not claim that a particular tool was used without supporting evidence.

5.4.3 Team Communication and Research Coordination

Team communication records can provide evidence of decisions made during development and integration. Where Microsoft Teams, meeting minutes, or other collaboration records were used, the report may summarise discussions concerning component dependencies, interface design, technical issues, testing decisions, and corrective actions.

The purpose of including these records is to demonstrate meaningful research coordination rather than simply the number of meetings or messages exchanged. Only relevant records should be selected, and personal or unrelated communication should be excluded.

Supporting meeting records may be included in the appendices and referenced from this section.

5.4.4 Source-Code and Implementation Evidence

Source-code and repository records provide evidence of development activity across the frontend, backend, and individual research components. The available VS Code screenshot shows a Git Graph containing commit and merge history, branch information, and a Source Control panel with pending changes.

This evidence supports the description of the project's version-control environment and recorded development history. Additional repository screenshots may be used to demonstrate the source-code structure or relevant implementation files, provided that they show the actual project.

Where available, commit history and repository records can also help identify changes associated with component implementation and integration. However, the report should not infer individual contributions solely from the number of commits. Contribution claims should be supported by relevant repository records and other evidence of the work performed.

5.4.5 Testing Records and Research Diaries

Testing records and research diaries can demonstrate how the team investigated technical problems, evaluated functionality, and refined the implementation. Relevant evidence may include test cases, expected and actual outputs, identified defects, corrective actions, and observations recorded during development.

The main chapter should summarise the most important findings from these records, while detailed test evidence can be included in Chapter 6 or the appendices. Research diaries can also help distinguish individual research contributions from collaborative development and integration activities.

Where records are incomplete or unavailable, the report should acknowledge this rather than imply that every development activity was formally documented.

5.4.6 Summary of Evidence and Supporting Appendices

The main chapter should present a concise summary of the strongest available evidence of team execution. Detailed screenshots, repository records, task summaries, and test outputs can be placed in the appendices and referenced from the relevant sections.

Evidence category	Examples of supporting records	Purpose
Project planning	Task board, task assignments, milestones	Demonstrates how work and responsibilities were organised.
Team coordination	Relevant meeting minutes and technical decision records	Demonstrates coordination and decisions affecting integration.
Source-code development	Git Graph, commit history, repository structure, relevant changes	Supports claims about version-control activity and implementation work.
Component testing	Test cases, expected outputs, actual outputs, and defect records	Demonstrates the evaluation of individual component functionality.
Integration testing	Interface tests and end-to-end workflow records	Supports claims about the combined operation of the system.
Research documentation	Research diaries, design records, and individual progress summaries	Demonstrates research activities and development decisions.
Deployment evidence	Verified configuration records and screenshots of the actual running application	Supports claims about the environment in which the application was executed.

Table 6:Project Execution Activities and Supporting Evidence

The evidence selected for the final report should be authentic, relevant, and consistent with the claims made in the chapter. Records that are unavailable should not be presented as completed evidence. Any limitations in the available development or testing documentation should also be acknowledged.

Overall, the implementation and execution evidence provides a transparent account of how the four research components were developed and brought together. When considered alongside the architecture and evaluation results presented in the other chapters, this evidence supports a clearer understanding of the technical progress and collaborative work undertaken by the research team.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

The evaluation of the integrated AI-driven tax advisory system focuses on determining whether the four components operate together as a coherent decision-support solution for personal income tax in Sri Lanka. While the individual theses examine the methods and performance of their respective components, the common report considers the complete workflow, the exchange of information between components, the consistency of system outputs, and the extent to which the integrated solution meets its overall objectives.

The evaluation framework is based on five main areas: functional correctness, integration reliability, tax-rule traceability, quality of AI-supported outputs, and responsible system operation. Functional correctness concerns whether the system performs the intended tasks. Integration reliability concerns whether information is transferred between components in a compatible format. Tax-rule traceability concerns whether tax calculations and explanations can be checked against the applicable legal provisions. AI-output quality concerns the relevance and reliability of transaction interpretations, advisory responses, and personalised recommendations. Responsible operation considers privacy, security, error handling, and the need for human verification.

The evaluation should distinguish between deterministic tax calculations and machine-learning predictions. Tax calculations are expected to follow the applicable rules for the selected year of assessment, whereas predictions and recommendations may involve uncertainty. This distinction is important because a prediction should not be treated as a legally authoritative tax result. Responsible AI guidance emphasises the need to assess risks and evaluate system behaviour in the context in which an AI system is used [4].

The following criteria provide a structure for evaluating the complete system.

Evaluation area	Evaluation criterion	Evidence or measure
Functional correctness	Each component performs its intended function	Test cases, expected outputs, and actual outputs
Component integration	Components exchange compatible information	Interface tests and request-response records
Tax calculation correctness	Calculations follow the applicable tax rules	Comparison against manually verified test cases and legal provisions
Transaction interpretation	Financial transactions are classified appropriately	Labelled test examples and classification evaluation results
Advisory response quality	Answers are relevant and supported by appropriate tax information	Representative questions, retrieved sources, and response reviews
Recommendation quality	Recommendations are relevant to the supplied taxpayer profile	Ranking evaluation, scenario testing, and review of outputs
Predictive impact analysis	Estimated outcomes follow the model assumptions	Simulation checks, scenario comparisons, and sensitivity analysis
Explainability	Relevant reasons for predictions or calculations can be inspected	Explanation outputs, rule references, and traceability records
Security and privacy	Sensitive information is handled according to defined requirements	Configuration review, security checks, and data-handling evidence

End-to-end reliability	The complete user workflow operates as intended	Recorded end-to-end test scenarios and identified failures

Table 7: Integrated Evaluation Criteria and Success Measures

The success criteria should be established before interpreting the results. For example, deterministic tax calculations should match independently verified expected results for the test cases used, while machine-learning components should be assessed using metrics suitable for their respective tasks. Classification, ranking, and prediction should not be evaluated using one common accuracy measure because each represents a different problem.

Where representative datasets, test records, or evaluation participants are unavailable, the report should state this limitation rather than imply that the corresponding criterion has been satisfied. Detailed component-level metrics should be reported in the individual theses, while this chapter should summarise the evidence relevant to the complete solution.

6.2 Integration and End-to-End Testing

Integration testing examines whether the four components can exchange information and contribute correctly to the complete tax advisory workflow. Testing only the components independently is insufficient because an output that is correct in isolation may still be incompatible with the next component or may be interpreted incorrectly when combined with other information.

The integration process considers the main relationships between the components. Component A, Transaction Semantic Reasoning, prepares structured information from financial transactions. Component B, Optimization and Explainable Engine, applies the relevant tax rules and produces calculation results with supporting explanations. Component C, Intelligent Tax Advisory Language Model, provides natural-language responses supported by relevant tax information. Component D, Personalized Recommendation and Predictive Impact Modeling Engine, uses relevant taxpayer and tax information to rank strategies and estimate potential financial effects.

The following testing areas are relevant to the integrated system.

6.2.1 Interface and data-exchange testing

Interface testing checks whether components receive the required inputs and return outputs in the expected format. This includes checking required fields, data types, valid values, error responses, and the handling of missing or incomplete information.

Particular attention should be given to the exchange of structured transaction information, tax calculation results, taxpayer profile information, retrieved tax sources, and recommendation outputs. A mismatch in field names, data types, or

assumptions could prevent a downstream component from processing an otherwise valid result.

Where an API connects the frontend and backend services, request and response behaviour should also be checked. API security guidance highlights the importance of appropriate validation and protection of interfaces that expose application functionality [19].

6.2.2 Tax-calculation and rule-traceability testing

Tax-calculation testing should verify whether the Optimization and Explainable Engine applies the appropriate rules for the selected year of assessment. Representative test cases should include different income situations, relevant reliefs, boundary conditions, and invalid or incomplete inputs where applicable.

Expected results should be independently calculated or otherwise verified against the relevant legal provisions before being compared with the system outputs. The evaluation should record the selected year of assessment, relevant input values, expected calculation, actual calculation, and any difference identified.

The traceability of tax explanations should also be examined. Where the system provides legal references, reviewers should verify whether the cited provisions support the associated explanation. This is important because retrieving a relevant-looking document does not, by itself, establish that the interpretation is legally correct.

6.2.3 AI advisory and retrieval testing

Testing of the Intelligent Tax Advisory Language Model should consider whether responses address the question, use relevant supporting information, and avoid presenting unsupported statements as established legal facts. Representative questions should cover common tax-related information needs, questions requiring clarification, and questions for which the available evidence is insufficient.

Where retrieval-augmented generation is used, the retrieved passages should be checked for relevance and suitability. The generated response should then be compared with the retrieved evidence to determine whether the explanation is adequately supported. Research on retrieval-augmented generation provides the methodological basis for combining retrieved knowledge with generated responses [7], while research on language-model limitations reinforces the need for independent checking [9].

6.2.4 Recommendation and predictive-impact testing

The Personalized Recommendation and Predictive Impact Modeling Engine should be evaluated using representative taxpayer profiles and scenarios appropriate to its intended functionality. Testing should examine whether recommendations reflect the

available profile information, whether the ranking process produces valid outputs, and whether the predicted effects are consistent with the assumptions used.

Where predictive models or ranking algorithms are used, the evaluation should apply suitable measures for the actual task and dataset. Explanations should also be reviewed to determine whether the factors presented as influential are consistent with the model outputs. If simulated future financial outcomes are produced, scenario and sensitivity checks should be used to examine how changes in assumptions affect the results [12], [15].

The evaluation must also acknowledge limitations in the available data. If adoption labels or taxpayer profiles are synthetic or proxy-based, the results cannot automatically be interpreted as evidence of how real taxpayers will behave.

6.2.5 End-to-end workflow testing

End-to-end testing examines the sequence of operations from the user's initial input to the final response. Representative workflows may include submitting financial information, reviewing transaction classifications, calculating tax, asking a related question, and exploring a personalised recommendation.

The tests should determine whether information remains consistent as it moves between components and whether the system handles invalid inputs or component failures appropriately. They should also examine whether users receive understandable error messages when a workflow cannot be completed.

Security, performance, usability, and user-acceptance testing should be included where such tests were actually conducted. If the group did not conduct formal penetration testing, load testing, or user studies, those activities should be identified as outstanding evaluation work rather than reported as completed tests.

6.3 Overall Results

The integrated results should bring together evidence from the individual component evaluations and the tests performed on the complete application. The purpose of this section is to establish what the group demonstrated at system level, rather than to repeat every algorithmic experiment described in the individual theses.

The overall results should be presented under the following areas, using the group's actual test records.

6.3.1 Functional and integration results

Functional results should indicate which of the four components were implemented and tested, whether the expected outputs were produced for the tested scenarios, and which issues remained unresolved. Integration results should show whether the

components exchanged information successfully and whether the combined workflows behaved as intended.

Evaluation area	Evidence to report	Result
Transaction interpretation	Number and type of test cases, expected and actual classifications	Insert verified result
Rule-based tax calculation	Number of verified calculation cases and discrepancies found	Insert verified result
Tax advisory responses	Questions evaluated, evidence checks, and identified response issues	Insert verified result
Personalised recommendations	Test profiles, ranking results, and observed limitations	Insert verified result
Predictive-impact modelling	Scenarios tested and consistency with stated assumptions	Insert verified result
Component integration	Interfaces and workflows successfully tested	Insert verified result
End-to-end operation	Completed workflows and unresolved failures	Insert verified result

Table 8:Summary of Integrated Functional and Testing Results

The table should be completed only from actual implementation and testing evidence. If a component was evaluated only through a limited set of demonstration examples, this should be stated clearly. A successful demonstration of one workflow does not establish that the system is reliable across all possible inputs.

6.3.2 Consistency and traceability of outputs

An important objective of the integrated solution is to preserve the relationship between financial information, tax calculations, supporting legal information, and recommendations. The evaluation should therefore examine whether downstream outputs can be traced to the information and assumptions on which they depend.

For example, a recommendation based on a tax calculation should not silently rely on an inconsistent assessment year or incomplete taxpayer profile. Similarly, a conversational explanation should not contradict the calculation engine without making the difference clear. Identified inconsistencies should be documented, together with the corrective actions taken or the limitations that remain.

Where the system uses explainability techniques, the results should indicate what those explanations establish and what they do not. An explanation of a model prediction can help users inspect the factors associated with an output, but it does not prove that the prediction is accurate or that the recommended strategy is legally suitable [12].

6.3.3 Interpretation of the integrated results

The overall findings should be interpreted in relation to the scope of the tests performed. If the recorded tests demonstrate that the components exchange information and complete selected workflows, the report may state that the integration was demonstrated for those scenarios. Broader claims about reliability, scalability, or suitability for real-world tax decisions require stronger evidence.

The group should also distinguish between implementation completion and evaluation success. A feature may be implemented but not fully validated. Likewise, a component may perform well on synthetic test data while its performance on real taxpayer information remains uncertain.

Consequently, the final assessment should identify the capabilities supported by recorded evidence, the areas that were only partially evaluated, and the additional testing required before wider use.

6.4 Stakeholder and User Validation

Stakeholder and user validation considers whether the integrated solution addresses the needs of its intended users and whether its outputs are understandable and useful in realistic situations. Technical correctness alone is not sufficient to establish usability or practical value, particularly in a domain where users may rely on information when making financial decisions.

Potential evaluation participants include individual taxpayers, tax professionals, accounting practitioners, or other stakeholders relevant to the project. The method of evaluation should be selected according to the participants available and the research objectives. Possible methods include structured task-based evaluation, questionnaires, interviews, expert review, and feedback on representative scenarios.

If stakeholder or user evaluation was conducted, this section should identify the participant group, number of participants, evaluation procedure, questions or tasks used, and method of analysing the collected evidence. It should summarise the principal findings, including aspects users found helpful, confusing, incomplete, or potentially misleading. Any changes made in response to the feedback should also be documented.

Evaluation element	Information to report
Participant group	Actual participant type and number
Evaluation method	Interview, questionnaire, task-based test, expert review, or another method actually used
Evaluation focus	Usability, clarity of explanations, relevance of recommendations, or other stated criteria
Main findings	Themes or results supported by the collected evidence
Changes made	Actual changes made following evaluation

Remaining concerns	Issues identified but not fully resolved

Table 9:Stakeholder and User Validation Summary

If no formal stakeholder or user evaluation was conducted, the report should state this explicitly. In that case, the system can be described as having undergone technical evaluation only to the extent supported by the available evidence. The need for future validation with representative taxpayers and qualified tax professionals should be acknowledged as a limitation.

Any stakeholder feedback should be reported accurately and without implying that a small or convenience-based group represents all Sri Lankan taxpayers.

6.5 Achievement of Common Objectives

The overall objectives of the project concern the development of an integrated solution that can interpret financial transactions, calculate tax using relevant rules, provide understandable tax information, and support personalised recommendations. Achievement should be assessed against evidence from the implementation and evaluation process rather than inferred solely from the existence of the four components

Common objective	Evidence required	Achievement status
Develop transaction interpretation and classification functionality	Implemented component and recorded classification tests	To be confirmed from test evidence
Apply year-specific tax rules and provide traceable calculations	Verified calculation cases and legal-rule checks	To be confirmed from test evidence
Provide tax-related responses supported by relevant information	Advisory test cases and source-verification records	To be confirmed from test evidence
Generate personalised tax-strategy recommendations	Tested taxpayer profiles and recommendation outputs	To be confirmed from test evidence
Estimate potential financial impacts under stated assumptions	Simulation tests and documented assumptions	To be confirmed from test evidence
Integrate the four components into a coherent application	Interface tests and end-to-end workflow records	To be confirmed from test evidence
Support transparency and responsible handling of information	Explanation checks and documented security and privacy measures	To be confirmed from available evidence

Table 10:A ssessment of Common Project Objectives

After reviewing the evidence, each objective should be marked as fully achieved, partly achieved, or not achieved. An objective should be considered fully achieved only when the implementation and evaluation evidence adequately supports the

intended outcome. Partial achievement may be appropriate where a feature has been implemented but only tested in a limited way. An objective should be marked as not achieved where the intended functionality was not implemented or the required outcome was not demonstrated.

The assessment should also explain any dependencies between objectives. For example, personalised recommendations may depend on the availability of reliable taxpayer information and correct tax calculations. If those upstream inputs are incomplete or uncertain, the quality of the downstream recommendation may also be affected.

This assessment provides a balanced account of the group's progress and helps identify which objectives have been demonstrated, which require further validation, and which remain future work.

6.6 Integrated Limitations and Threats

Although the integrated system brings together four complementary components, several limitations may affect the reliability and generalisability of its outputs. These limitations should be considered when interpreting the evaluation results and determining whether the system is suitable for use beyond the research environment.

6.6.1 Data quality and representativeness

The performance of transaction interpretation and predictive models depends partly on the quality, completeness, and representativeness of the data used. Financial transaction descriptions may be ambiguous, while synthetic taxpayer profiles may not capture the full diversity of real financial circumstances. Where synthetic data or proxy labels are used, measured performance may not generalise to real taxpayers.

The group should therefore document the origin and characteristics of the data used, the limitations of any synthetic datasets, and the steps taken to avoid drawing conclusions beyond the available evidence.

6.6.2 Tax-rule currency and legal interpretation

Tax legislation can change, and a rule that applies to one year of assessment may not apply to another. The system's usefulness therefore depends on maintaining accurate tax rules, identifying the relevant assessment year, and validating calculations against authoritative legal sources.

Retrieval-based advisory responses also carry a risk of selecting outdated, incomplete, or insufficiently relevant legal information. Even when a source is retrieved correctly, the generated explanation may misinterpret its meaning. Continued legal review and systematic validation are necessary before the system can be relied upon for consequential tax decisions [1], [7].

6.6.3 Dependencies between components

The components depend on one another's inputs and outputs. Incorrect transaction classifications can affect tax calculations, incomplete taxpayer profiles can reduce the relevance of recommendations, and unsupported advisory responses can create confusion about the basis of a result.

These dependencies mean that testing each component independently cannot establish the reliability of the complete application. End-to-end testing should consider how errors propagate through the workflow and whether the system communicates uncertainty or failure clearly.

6.6.4 Model explainability and predictive uncertainty

Explainability methods can help users understand aspects of model behaviour, but they do not establish legal correctness or guarantee future outcomes [5], [12]. Recommendation rankings and adoption estimates are influenced by the data, assumptions, and modelling choices used. If the system relies on synthetic or proxy-based behaviour labels, adoption predictions should be interpreted as experimental estimates rather than verified descriptions of real taxpayer behaviour.

Similarly, financial-impact simulations depend on assumptions about future conditions. Monte Carlo methods can explore uncertainty across specified scenarios, but their outputs are only as meaningful as the assumptions and distributions used in the simulation [15]. They should not be interpreted as guaranteed financial outcomes.

6.6.5 Evaluation and generalisability

The strength of the evaluation depends on the number, diversity, and quality of test cases. A limited collection of examples may demonstrate that a workflow functions under selected conditions, but it may not reveal failures that occur with unusual inputs, different taxpayer circumstances, or changes in legislation.

If formal user studies, professional reviews, security assessments, or performance tests were not conducted, the report should identify these omissions. Their absence limits the conclusions that can be drawn about usability, security, scalability, and practical acceptance.

6.6.6 Privacy, security, and operational limitations

The system handles information that may be personal or financially sensitive. Its responsible use therefore requires appropriate data handling, access controls, secure API communication, and careful management of credentials and external services. Personal-data protection requirements and recognised API security guidance provide relevant considerations for these responsibilities [18], [19].

The extent to which these protections have been implemented and tested should be reported accurately. A design intention or configuration setting alone does not establish that the system is secure. Further security review may be required before deployment in a real operational environment.

6.6.7 Overall implications of the limitations

The limitations described above do not remove the value of investigating an integrated AI-driven tax advisory system. Instead, they define the conditions under which the results should be interpreted. The current evaluation should establish what the group has demonstrated through actual implementation and testing, while distinguishing these findings from capabilities that remain unverified.

Future work should prioritise broader and more representative testing, regular verification of tax rules, improved evaluation of model outputs, professional and user validation, and stronger evidence of privacy and security controls. These steps would provide a firmer basis for assessing whether the integrated system can be used reliably in practical tax advisory scenarios

7. COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

7.1.1 Value Proposition of the Integrated System

The proposed AI-driven tax advisory system aims to make personal income tax information, calculations, and decision-support capabilities more accessible to individuals in Sri Lanka. Traditional tax-related tasks may require users to interpret financial records, understand applicable tax rules, calculate their tax obligations, and identify appropriate financial strategies. These activities can become difficult when users have limited tax knowledge or must consider changes in tax legislation.

The proposed system addresses these challenges through four integrated components. Component A interprets financial transactions, Component B performs tax calculations and provides explanations, Component C supports users through tax-related advisory responses, and Component D generates personalised recommendations and estimates their potential financial impact.

The main value proposition is the combination of these functions within a common application. Instead of treating transaction interpretation, tax calculation, advisory support, and financial recommendations as entirely separate activities, the system provides a structure through which they can support one another.

The potential value of the system lies in helping users organise relevant information, understand how tax-related results are produced, and explore possible strategies. However, the extent to which these benefits are achieved must be established

through usability testing, calculation validation, stakeholder feedback, and other appropriate evaluation activities.

7.1.2 Primary Beneficiaries and Their Expected Benefits

The system is intended to support several groups of users who may benefit from improved access to tax-related information and decision support.

Individual taxpayers

Individual taxpayers are the primary intended beneficiaries. The system aims to help them interpret relevant financial information, understand tax calculations, ask tax-related questions, and examine potential recommendations. These functions may be particularly useful for individuals who find tax terminology or calculation procedures difficult to understand.

Accountants and tax professionals

Accountants and tax professionals may use the system as a supporting tool for organising information, reviewing calculations, and exploring potential tax-related strategies. The system could assist with preliminary analysis, while professional judgement would remain important when interpreting complex circumstances or confirming legal obligations.

Small businesses and self-employed individuals

Although the system focuses on personal income tax advisory, some of its capabilities may be relevant to eligible self-employed individuals and people managing multiple sources of personal income. Its usefulness for these groups would depend on the financial information and tax scenarios supported by the implemented system.

Educational and research users

Students, researchers, and other educational users may benefit from the system's explanations of tax rules and calculations. The application may provide a practical environment for exploring the relationship between financial information, tax legislation, and AI-assisted decision support.

7.1.3 Wider Social and Economic Value

Beyond direct users, the proposed system has the potential to contribute to improved access to understandable tax information. Clear explanations and accessible digital services may help individuals develop a better understanding of their tax obligations and the reasoning behind tax-related calculations.

The integration of structured tax rules with AI-based advisory and recommendation functions also demonstrates how digital technologies can be applied to a specialised public-interest domain. The approach may provide a foundation for future research into explainable decision-support systems for financial and administrative tasks.

Nevertheless, these wider benefits should be regarded as potential outcomes rather than established impacts. Demonstrating meaningful social or economic value would require evidence of user adoption, improved understanding, reliable calculations, and measurable improvements in the relevant tax-related tasks.

7.2 Market Alternatives and Differentiation

7.2.1 Target Market and Intended User Segments

The initial target market for the proposed system is individuals who need assistance with personal income tax calculations, explanations, and related financial decisions in Sri Lanka. Within this group, potential users include salaried employees, individuals with multiple income sources, and people who require support in understanding applicable tax rules.

A secondary target segment includes accountants, tax consultants, and other professionals who may benefit from tools that assist with preliminary analysis and the explanation of tax-related results.

The potential relevance of the system to these groups does not establish that they are willing to adopt or pay for it. Market demand, preferred features, acceptable pricing, and willingness to trust AI-assisted tax guidance would need to be investigated through interviews, surveys, prototype demonstrations, or pilot studies.

7.2.2 Existing Alternatives and Substitute Solutions

Users already have several ways to obtain tax information and assistance. These alternatives provide a useful basis for evaluating the proposed system's intended position.

Official tax information and guidance

The Inland Revenue Department of Sri Lanka provides official information on income tax and related legislation [1]. These sources remain essential for identifying applicable legal requirements. However, individual users may still need to interpret the relevant provisions and apply them to their own financial circumstances.

Manual calculations and spreadsheets

Individuals and professionals may use spreadsheets, calculators, and manual procedures to organise financial information and estimate tax liabilities. These approaches can be flexible but may require users to understand the relevant rules and maintain their calculations when circumstances or legislation change.

Accountants and tax consultants

Professional services provide personalised interpretation and support for tax-related matters. Their advice can be especially important in complex situations. The proposed system is not intended to replace professional services; rather, it may provide preliminary information and decision support for suitable cases.

General-purpose AI assistants and digital advisory tools

General-purpose AI assistants can explain tax concepts and answer questions, while specialised digital tools may support particular calculation or filing tasks. However, their suitability depends on the accuracy of their information, their coverage of Sri Lankan tax rules, and their ability to account for individual circumstances.

These alternatives should not be assumed to have the same capabilities or limitations. A detailed competitive assessment would require direct comparison of their supported functions, legal coverage, costs, data-handling practices, and user experience.

7.2.3 Differentiation of the Proposed System

The proposed system's intended differentiation lies in its integration of four complementary capabilities within a Sri Lankan personal income tax context.

First, transaction interpretation provides a structured way to convert financial records into information that can support tax-related processing. Second, the Optimization and Explainable Engine applies relevant tax rules and provides calculation explanations. Third, the Intelligent Tax Advisory Language Model supports tax-related questions. Fourth, the Personalized Recommendation and Predictive Impact Modeling Engine explores recommendations and estimates their potential financial impact.

The combination of these functions may provide a more connected decision-support experience than approaches that focus only on information retrieval or isolated calculations. The use of structured tax rules and explanations is particularly relevant because tax-related outputs should be traceable to appropriate rules and assumptions.

However, these features do not establish that the proposed system is more accurate, less expensive, or more effective than existing alternatives. Such claims would require comparative testing and evidence from representative users and use cases.

7.2.4 Competitive Position and Future Differentiation

To establish a defensible position, future development should focus on maintaining accurate Sri Lankan tax-rule representations, improving the clarity of explanations, validating transaction classifications, and demonstrating the reliability of personalised recommendations.

Other areas for differentiation may include a straightforward user interface, transparent assumptions, appropriate privacy protections, and the ability to trace tax-related outputs to supporting information.

A future market assessment should compare the system with relevant official resources, manual methods, professional services, and available digital alternatives. The comparison should consider functionality, accuracy, ease of use, maintenance requirements, user trust, and the cost of providing the service.

At the present research stage, the system's differentiation should therefore be described as a proposed technical and functional contribution rather than a proven competitive advantage.

7.3 Business Model and Feasibility

7.3.1 Potential Business and Sustainability Models

The integrated tax advisory system could support several possible operating models if it is developed beyond the research prototype. The most suitable model would depend on user demand, operational costs, regulatory requirements, and the level of assistance that users require.

Free access supported by institutional or project funding

The system could be made available as an educational or public-interest resource through university, research, or institutional support. This approach could prioritise accessibility while funding ongoing maintenance through grants, partnerships, or other appropriate sources. Its long-term feasibility would depend on securing sufficient and continuing support.

Freemium access

A freemium model could provide basic tax-information and calculation features at no charge while reserving advanced analysis or additional services for a paid tier. This model would require evidence that users value the additional functions enough to support the associated costs.

Subscription or professional-use model

A subscription model could be considered for accountants, consultants, or other professional users who require repeated access to supported tax-related functions. Before adopting this approach, the system would need to demonstrate reliable operation, appropriate legal coverage, secure handling of information, and a clear benefit to the intended users.

These models are potential future options, not confirmed commercial arrangements. Selecting one would require further stakeholder research and financial analysis.

7.3.2 Channels, Partnerships, and Resources

Potential distribution channels include a web-based application, educational demonstrations, professional networks, and collaboration with institutions interested in digital tax support. These channels could help users access the system and provide feedback during future pilot studies.

Potential partners may include universities, tax professionals, financial-technology organisations, or relevant institutions. Such partnerships could support domain validation, usability testing, technical development, and the review of tax-related information. No partnership should be represented as established unless a formal or otherwise verifiable arrangement exists.

The main resources required for continued development include the application source code, the four component implementations, the supporting tax-rule information, computing resources, and people with expertise in software engineering, taxation, AI, and information security.

Maintaining the tax-rule information is particularly important because legislative changes may require updates to the relevant rules and renewed validation of affected calculations. Continued operation would therefore require both technical maintenance and appropriate tax-domain review.

7.3.3 Cost Structure and Financial Considerations

The major potential costs associated with operating the system include software development and maintenance, computing infrastructure, external AI services where used, information storage, security measures, and the review of tax-rule updates.

Additional costs may arise from user support, testing, documentation, compliance reviews, and the investigation of errors. The extent of these costs would depend on the architecture, number of users, frequency of use, and services selected for deployment.

A reliable financial assessment should estimate both initial development costs and recurring operational expenses. It should also consider the cost of keeping tax-related information current and validating changes before they are used in calculations or advisory responses.

Since confirmed operating costs, revenue, customer numbers, and willingness-to-pay evidence have not been established in the information available for this report, a numerical profitability claim would not be justified. These values should be investigated through a future feasibility study.

7.3.4 Technical and Operational Feasibility

The project demonstrates a technical approach for combining a web interface, backend processing, structured tax rules, and AI-based functions within a common application architecture. The identified use of React, TypeScript, Vite, and FastAPI provides a foundation for implementing and integrating the system's functionality.

However, technical feasibility involves more than selecting suitable technologies. It also depends on the correctness of the implemented tax rules, compatibility between components, response times, security, maintainability, and the ability to recover from operational problems.

Operational feasibility additionally depends on the availability of suitable technical support, the ability to maintain assessment-year-specific information, and the establishment of procedures for handling errors and user enquiries.

The current research prototype can provide a basis for investigating these requirements. Wider deployment would require further evidence from integration testing, security assessment, usability evaluation, operational trials, and validation by appropriately qualified tax professionals.

7.3.5 Future Validation of Commercial Feasibility

Commercial feasibility should be investigated through a structured process rather than assumed from technical implementation alone. An initial pilot could examine whether intended users understand the interface, find the explanations useful, and can complete representative tax-related tasks accurately.

Subsequent evaluation could measure user satisfaction, task completion, calculation correctness, operating costs, and willingness to adopt or pay for the service. Feedback from tax professionals could also help identify missing functionality and cases requiring expert review.

The findings could then inform decisions about the most suitable operating model, the resources needed for continued maintenance, and whether the system should remain a research or educational resource or be developed into a broader service.

7.4 Adoption, Intellectual Property, and Compliance

7.4.1 Adoption Barriers and User Trust

The adoption of an AI-driven tax advisory system may be affected by several factors. Users may be concerned about the accuracy of tax calculations, the reliability of AI-generated responses, the privacy of their financial information, and the difficulty of understanding technical explanations.

These concerns are especially important in taxation because incorrect information may lead to inappropriate financial decisions or compliance errors. Users must be

able to understand the limitations of the system and recognise when a result requires additional verification.

The system should therefore present tax calculations with clear explanations, distinguish estimates from confirmed amounts, and communicate uncertainty where relevant. Demonstrations and pilot testing may also help identify usability problems and determine whether users understand the information presented.

Trust should be developed through verifiable accuracy, transparent limitations, and appropriate data-handling practices rather than through claims that AI-generated results are always correct.

7.4.2 Intellectual Property and Software Licensing

The project includes software code, component designs, structured tax-rule representations, research documentation, and other development artefacts. The ownership and permitted use of these materials should be established in accordance with the university's research policies, the group's agreements, and the licences of any third-party software or resources used.

Before distributing or commercialising the system, the team should review the licences of its frameworks, libraries, AI services, and external datasets. The use of open-source software does not automatically grant unrestricted rights to all related materials, and the conditions of individual licences must be respected.

The team should also clarify ownership and permission arrangements for jointly developed code and research outputs. The report should not claim that a patent, trademark, copyright registration, or commercial licence has been obtained unless the relevant action has actually taken place.

7.4.3 Data Protection and Privacy Responsibilities

The system may process financial records and personal information that require careful protection. Responsible operation therefore requires consideration of how information is collected, used, transmitted, stored, and deleted.

The system should collect only the information necessary for its intended functions and restrict access according to legitimate operational requirements. Sensitive credentials should be protected, and personal information should not be unnecessarily exposed in application logs, screenshots, source-code repositories, or public demonstrations.

Where external AI services are used, the team should examine their data-handling conditions and determine whether submitted information is processed or retained by the service provider. Appropriate measures should be taken to avoid sending unnecessary sensitive information to external services.

The project should be reviewed against applicable Sri Lankan data-protection requirements, including the Personal Data Protection Act, No. 9 of 2022, and any amendments or provisions in force at the time of deployment. The precise legal obligations should be confirmed through current official sources and appropriate legal guidance rather than assumed from the prototype's research scope.

7.4.4 Tax Compliance and Professional Responsibility

Tax calculations and advisory responses must be consistent with the relevant legislation and the assessment year being considered. The system should distinguish between information obtained from authoritative sources, structured rules implemented in the application, and generated or predicted outputs.

Tax rules should be checked against relevant official sources, including the Inland Revenue Department of Sri Lanka [1]. Changes to legislation should be reviewed before the corresponding rules are updated, and affected calculations should be tested to reduce the risk of introducing errors.

The system should be positioned as a decision-support tool rather than an unconditional substitute for professional tax advice. Where the available information is incomplete, a legal provision is ambiguous, or a taxpayer's circumstances require specialist interpretation, the system should communicate the limitation and encourage appropriate verification.

Professional responsibility also includes avoiding misleading claims about the system's capabilities. In particular, predictive estimates should not be presented as guaranteed savings, and a generated explanation should not be treated as proof that the underlying calculation is legally correct.

7.4.5 Responsible AI, Transparency, and Safety

Responsible AI is important because the system combines financial information, rule-based calculations, language-model responses, and predictive recommendations. Each function introduces different risks and should be evaluated according to its purpose.

The tax calculation functionality should be checked against suitable reference cases. Advisory responses should be assessed for factual support and relevance. Predictive recommendations should be evaluated using appropriate measures and should clearly communicate the assumptions and limitations underlying their estimates.

Where machine-learning models are trained using synthetic data or proxy labels, this limitation should be disclosed. Performance on synthetic or simulated examples does not establish equivalent performance for real taxpayers in practical use.

The system should also avoid encouraging users to rely on unsupported deductions, unverified tax strategies, or recommendations that conflict with applicable

legislation. Appropriate validation, clear explanations, and access to supporting information can help reduce these risks.

These considerations are consistent with responsible AI risk-management principles that emphasise reliability, transparency, accountability, and the evaluation of potential harm [4], [5].

7.5 Sustainable Development Goal Contribution

7.5.1 Contribution to Sustainable Development Goal 9

Sustainable Development Goal 9 focuses on industry, innovation, and infrastructure [6]. The proposed system contributes to this goal at a research and technological development level by exploring how AI, structured tax rules, and web-based software can be combined to address a practical financial-administration problem.

The four-component architecture demonstrates an approach to integrating transaction interpretation, explainable tax calculations, advisory responses, and predictive recommendations. This work may provide a foundation for further digital innovation in tax-related decision support.

However, the existence of a prototype does not establish a measurable improvement in national digital infrastructure or industrial productivity. A stronger claim would require evidence of deployment, sustained use, and measurable benefits for the intended users.

7.5.2 Contribution to Sustainable Development Goal 10

Sustainable Development Goal 10 focuses on reducing inequality within and among countries [6]. The proposed system may contribute indirectly by making tax-related information and calculation explanations easier to access for individuals who find existing procedures difficult to understand.

A clear and accessible interface may help users with different levels of tax knowledge engage with relevant information. The integration of transaction processing, calculations, and advisory support may also reduce some of the effort required to organise information and understand tax-related results.

Nevertheless, digital accessibility alone does not guarantee equal access. Users may face barriers related to internet availability, digital literacy, language, disability, or access to suitable devices. Future development should consider these barriers and investigate whether the system is equally understandable and useful for different user groups.

Any claim that the system reduces inequality would require evidence from user studies or deployment data showing that it improves access or outcomes for the intended beneficiaries.

7.5.3 Contribution to Sustainable Development Goal 16

Sustainable Development Goal 16 focuses on peace, justice, and strong institutions [6]. The proposed system may support aspects of this goal by making tax-related information more understandable and by providing explanations for calculations and recommendations.

Traceable tax rules and evidence-supported advisory responses may help users review the basis of the information they receive. The system's focus on explainability also encourages attention to transparency and accountability when AI technologies are applied to tax-related decisions.

However, the system is not an official government tax platform, and its development does not independently demonstrate improvements in institutional transparency or public accountability. Its contribution should be understood as a potential supporting role that depends on accuracy, responsible use, and appropriate validation.

7.5.4 Potential Negative Effects and Risk Mitigation

Although the system may provide useful decision support, it also presents potential risks that could undermine its intended benefits.

Incorrect tax rules or calculations could mislead users. Unsupported AI-generated responses could create false confidence, while inaccurate predictive estimates could influence financial decisions. Privacy weaknesses could expose sensitive financial information, and limited accessibility could prevent some intended users from benefiting from the application.

These risks should be addressed through appropriate measures, including:

- validating calculations against relevant tax rules and reference cases;
- reviewing tax-rule updates against authoritative legislative sources;
- providing explanations and communicating uncertainty where appropriate;
- distinguishing synthetic-data evaluation from validation using real-world cases;
- protecting financial information and limiting unnecessary data collection;
- evaluating accessibility and usability with representative users; and
- encouraging professional verification when cases are complex or the available information is insufficient.

The effectiveness of these measures should be assessed through testing and documented evidence. Their inclusion as design principles does not establish that all risks have already been eliminated.

7.5.5 Long-Term Sustainability and Future Impact Assessment

The long-term contribution of the system depends on maintaining the accuracy, usefulness, accessibility, and security of the application after the initial research period. Tax legislation changes over time, and the system would require a reliable process for reviewing and updating relevant rules and supporting knowledge.

Continued sustainability would also depend on maintaining the software, monitoring its behaviour, addressing reported problems, and ensuring that its outputs remain appropriate for the intended users. Any future deployment would require suitable technical resources and clear responsibility for maintaining the system.

The project's contribution to the selected Sustainable Development Goals should be evaluated using measurable indicators where possible. Potential indicators include calculation correctness, user task-completion rates, user understanding of tax explanations, accessibility across user groups, and the proportion of advisory responses supported by appropriate evidence.

Such indicators would help establish whether the system delivers practical benefits rather than relying only on its intended purpose. Until these outcomes have been measured, the contribution to SDGs 9, 10, and 16 should be presented as plausible and conditional on successful validation and responsible deployment.

8. CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

8.1.1 Addressing the Overall Research Problem

Personal income tax compliance requires individuals to organise financial information, understand applicable tax legislation, calculate their tax obligations, and make informed decisions based on their financial circumstances. These activities can be challenging when financial records require interpretation, tax rules vary across assessment years, or users have difficulty understanding technical tax information. Addressing these challenges requires more than an isolated calculation tool or a general source of tax-related information.

This research addressed the problem by developing an integrated AI-driven tax advisory system designed to support personal income tax compliance and advisory in Sri Lanka. The proposed solution brings together transaction interpretation, tax calculation and explanation, conversational tax advisory, and personalised recommendation capabilities within a common application architecture.

The system is organised into four complementary components. Component A, Transaction Semantic Reasoning, focuses on interpreting and classifying financial transactions. Component B, Optimization and Explainable Engine, applies relevant tax rules and provides calculation explanations. Component C, Intelligent Tax

Advisory Language Model, supports users in obtaining understandable tax-related guidance. Component D, Personalized Recommendation and Predictive Impact Modeling Engine, explores personalised recommendations and estimates their potential financial impact.

The integration of these components provides a structured approach to addressing different stages of the tax-related decision-making process. Financial transaction information can support tax calculations, calculation results can contribute to personalised analysis, and advisory functionality can help users understand relevant tax concepts. Together, these capabilities establish a foundation for a more connected tax decision-support experience.

8.1.2 Overall Findings and Research Significance

The principal outcome of the research is the design and implementation of a multi-component approach that combines structured tax-rule processing with AI-based interpretation, advisory, and recommendation functionality. The common architecture provides a way to organise these capabilities while maintaining distinct responsibilities for the individual components.

An important aspect of the proposed approach is the emphasis on understandable and reviewable outputs. Tax calculations should be supported by relevant rules and explanations, advisory responses should be grounded in appropriate information, and predictive recommendations should communicate their assumptions and limitations. These considerations are important because tax-related decisions may have financial and compliance consequences for users.

The research also demonstrates the importance of treating integration as a central design consideration. The usefulness of the overall system depends not only on the functionality of each component but also on the consistency of the information exchanged between them. Incorrect transaction classifications, outdated tax rules, unsupported advisory responses, or unrealistic predictive assumptions could affect the reliability of downstream results.

The significance of the research therefore lies in its integrated problem-solving approach and its consideration of explainability, tax-rule consistency, and personalised decision support. The extent to which the system improves accuracy, usability, or user decision-making must be established through the evaluation evidence presented in Chapter 6. These benefits should not be assumed solely from the system's architecture or implemented features.

8.1.3 Overall Limitations and Concluding Remarks

Despite its potential, the proposed system has limitations that must be considered when interpreting its contribution. Tax-related calculations depend on the correct representation of applicable legislation and assessment-year-specific rules. AI-generated advisory responses may require further verification, while personalised

recommendations depend on the quality of the available information and the assumptions used in predictive analysis.

Where synthetic data or proxy labels are used, the resulting evaluation may not fully represent the behaviour of real taxpayers or the conditions encountered in practical use. Similarly, successful execution in a development environment does not, by itself, establish that the system is ready for production deployment or can support large numbers of users.

These limitations highlight the need for careful validation, appropriate human oversight, and continued maintenance. The system should be regarded as a decision-support solution rather than an unconditional replacement for official tax guidance or professional judgement.

In conclusion, the research establishes an integrated approach to supporting personal income tax compliance and advisory in Sri Lanka through four complementary AI-driven components. Its practical value will depend on the correctness of its tax-related outputs, the reliability of its integration, and its ability to provide useful and understandable assistance to intended users. Further evaluation and refinement are necessary to determine the extent of its benefits in real-world settings.

8.2 Principal Integrated Contribution

8.2.1 Integration of Four Complementary Research Components

The principal integrated contribution of this research is the combination of four distinct research components into a common tax advisory system. Each component addresses a particular problem, but the overall architecture connects their outputs to support a broader decision-making process.

Component A contributes the interpretation of financial transactions, Component B contributes tax calculations and explanations, Component C contributes conversational tax advisory, and Component D contributes personalised recommendations and predictive financial impact analysis. Their combined structure provides a broader scope of functionality than a solution focused exclusively on one of these tasks.

This integration also establishes clear functional relationships between transaction information, tax-rule application, advisory knowledge, and personalised analysis. These relationships provide a foundation for extending the system while maintaining separation between component responsibilities.

8.2.2 A Common Architecture for Tax-Related Decision Support

A second contribution is the common application architecture that connects the frontend, backend communication layer, component functionality, and supporting information resources. The use of a common web interface and backend API

structure provides a consistent way for users to access the different functions of the system.

The architecture also highlights the importance of shared information formats and defined interfaces. For example, structured transaction information must be suitable for tax processing, and calculation results must retain enough context to support downstream recommendation functions. Such dependencies demonstrate why integration and interface validation are essential when developing multi-component decision-support systems.

The architecture provides a basis for further development, testing, and maintenance. Its long-term effectiveness will depend on how well the implemented interfaces, error-handling mechanisms, and shared data models support reliable operation.

8.2.3 Combining Explainability, Advisory Support, and Personalisation

The research also contributes an approach to bringing explainable calculations, conversational advisory, and personalised analysis into one system. These functions address different user needs: understanding how a tax-related result was obtained, obtaining guidance about relevant tax concepts, and exploring the possible consequences of different financial strategies.

Combining these capabilities may help users move from reviewing financial information towards understanding tax-related results and considering possible actions. However, the quality of this experience depends on the reliability of the underlying rules, the relevance of advisory information, and the validity of the assumptions used in recommendations.

The integrated approach therefore provides a basis for investigating how different AI and rule-based techniques can complement one another in a specialised financial-advisory context.

8.2.4 Contribution to Future Research and Development

Beyond the current implementation, the project provides a foundation for future research into AI-supported tax compliance, explainable financial decision support, and personalised advisory systems. The component-based structure allows future researchers to evaluate individual functions while also examining the effects of integrating them into a common application.

The research also identifies important areas for further investigation, including assessment-year-specific rule maintenance, the reliability of generated tax guidance, the evaluation of predictive recommendations, and the protection of sensitive financial information.

The principal contribution should therefore be understood as an integrated research and development outcome rather than proof of a fully validated commercial product.

Its broader value will depend on further testing, external review, and evidence that the system provides reliable and useful support for real users.

8.3 Future Directions

8.3.1 Strengthening Evaluation Through Representative Test Data

A high priority for future work is to evaluate the system using a broader collection of representative tax scenarios. These should include different income sources, relevant reliefs, varying assessment years, incomplete information, and cases that require additional clarification.

For Component A, evaluation should measure the quality of transaction classifications against appropriately labelled reference data. For Component B, calculated results should be compared with verified reference calculations based on the applicable tax rules. For Component C, advisory responses should be assessed for factual correctness, relevance, clarity, and support from authoritative sources. For Component D, recommendation quality and predictive estimates should be evaluated using suitable metrics and clearly defined assumptions.

Where synthetic data is used, its role and limitations should be documented. Future work should investigate the availability of suitable real-world or anonymised evaluation data, subject to appropriate permissions and privacy safeguards.

8.3.2 Improving Tax-Rule Maintenance and Legal Validation

The system's reliability depends on the accuracy and currency of the tax rules used by its calculation and advisory functions. Future development should establish a systematic process for reviewing relevant legislative changes, updating structured rules, and verifying their effects on existing functionality.

Each rule update should be linked to an authoritative source and the relevant assessment year. Affected calculation scenarios should be retested before updated rules are relied upon. Where a legal provision is ambiguous or requires interpretation, the system should support review by an appropriately qualified tax professional.

This work would help reduce the risk of outdated or inconsistent tax information and provide a clearer basis for maintaining the system over time.

8.3.3 Enhancing the Reliability and Explainability of AI Outputs

Further work should focus on improving the reliability of transaction interpretation, tax-related advisory responses, and personalised recommendations. For the advisory component, this may include strengthening retrieval of supporting information, improving source references, and evaluating responses against carefully prepared questions and reference answers.

For the recommendation component, future research should assess the stability of rankings and the sensitivity of predicted financial impacts to changes in assumptions. Where synthetic data or proxy labels are used, further validation is needed to determine how well the resulting models generalise beyond the training data.

Explanations should make it easier for users to understand the factors influencing a result without implying that a model-based output is guaranteed to be correct. The system should also communicate when the available information is insufficient for a reliable conclusion.

8.3.4 Conducting User Studies and Professional Validation

Future development should include structured evaluations involving intended users, including individual taxpayers and, where feasible, accountants or tax professionals. These evaluations could examine whether users understand the interface, can complete representative tasks, interpret the explanations correctly, and recognise the limitations of AI-generated outputs.

Usability studies could measure task-completion rates, time required to complete selected tasks, user-reported clarity, and common sources of confusion. Professional review could assess the relevance of tax calculations, the quality of advisory responses, and the suitability of recommendations for the supported scenarios.

The findings should be used to prioritise improvements rather than assuming that the availability of multiple functions automatically results in a better user experience.

8.3.5 Strengthening Security, Privacy, and Responsible Operation

Before wider deployment, the system should undergo a structured review of its security and privacy practices. Areas for investigation include access control, secure configuration, handling of credentials, data minimisation, protection of stored information, and the treatment of data submitted to external services.

Future work should also establish clear procedures for handling errors, responding to reported problems, and maintaining tax-related information. Users should be informed about the purpose of data collection, the limitations of the system, and the circumstances in which professional verification may be necessary.

These measures would support responsible operation and help build user confidence while reducing the risks associated with processing sensitive financial information.

8.3.6 Evaluating Deployment and Operational Feasibility

The current research provides a foundation for further evaluation of deployment requirements. Future work should assess the infrastructure needed to operate the integrated application, the cost of external services where applicable, response times under representative workloads, and the resources required to maintain the system.

Load testing could investigate performance under increasing numbers of requests, while reliability testing could examine how the application behaves when a component or supporting resource becomes unavailable. Backup and restoration procedures should also be established and tested where persistent data or configuration resources require recovery.

A limited pilot deployment could then be used to gather evidence about operational behaviour, user needs, maintenance requirements, and potential adoption barriers. Broader deployment should be considered only after the relevant technical, legal, security, and usability requirements have been addressed.

8.3.7 Expanding Accessibility and Future Functionality

Further development could improve accessibility through clearer language, more understandable explanations, and interface design suitable for users with different levels of digital and tax knowledge. Additional language support may also be worth investigating to improve accessibility for users who are more comfortable receiving guidance in Sinhala or Tamil, subject to the system's ability to provide accurate tax-related information in those languages.

Other possible improvements include broader support for relevant tax scenarios, more detailed explanations of recommendation assumptions, and additional tools for reviewing the effects of alternative financial circumstances.

Any new functionality should be prioritised according to demonstrated user needs, implementation feasibility, and the availability of reliable information. Expansion should not compromise the correctness, clarity, or maintainability of the existing system.

8.3.8 Overall Future Development Priorities

The future development of the integrated tax advisory system should follow a staged approach. The immediate priority is to strengthen the evidence supporting the correctness and reliability of the four components. The next priority is to improve user understanding and validate the integrated workflows through representative testing and professional review. Security, privacy, maintainability, and operational feasibility should then be addressed as part of preparing for wider evaluation or deployment.

Commercialisation and broader adoption should remain dependent on evidence of practical value, user acceptance, and sustainable operating arrangements. By following these priorities, future work can build on the current research while addressing the technical and operational limitations that must be resolved before the system can be relied upon in real-world tax-related decision-making.

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APPENDICES

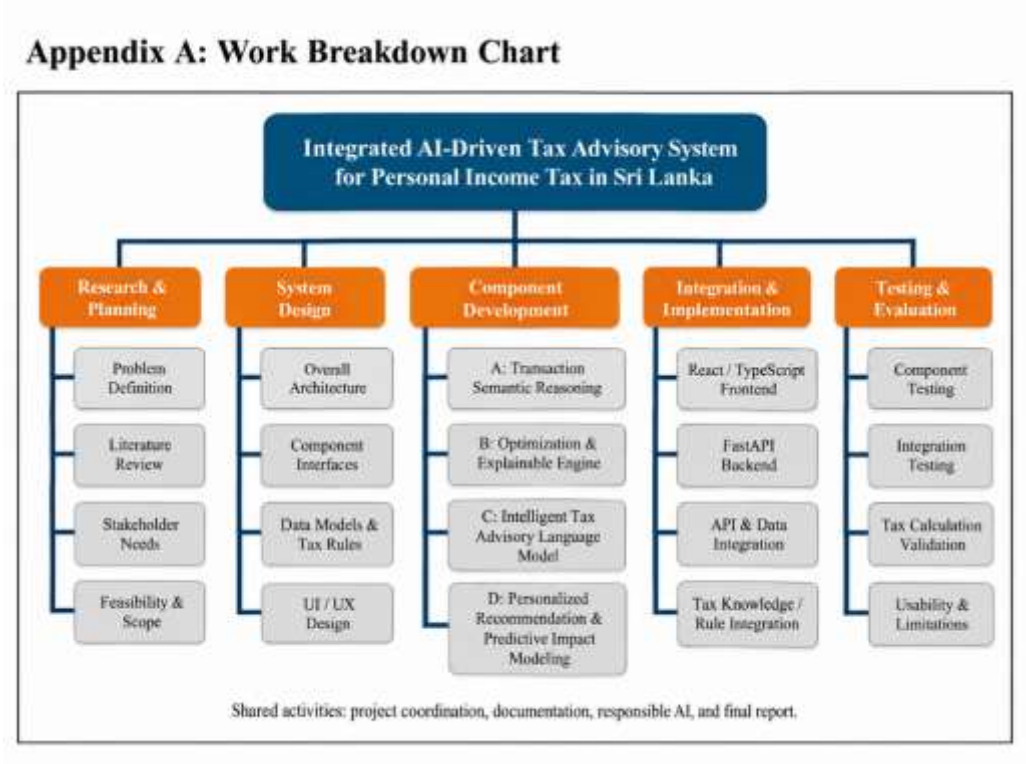


Figure 3: Work Breakdown Structure of the Integrated AI-Driven Tax Advisory System